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PROJECTION-TRANSPORT STEERING: CONDITIONAL CONTROL OF REASONING BEHAVIOR IN LLMS

Abstract. Activation steering promises behavior control without retraining, but a global shift of the residual stream suppresses useful and harmful behavior together. In this work we treat steering as an optimal-transport problem on a low-dimensional behavioral subspace and separate the decision to intervene from the intervention itself. We prove that the transport map between projection laws is the unique minimum-displacement intervention local to that subspace, that the selectivity of any transfer function on an axis is capped by how well the axis separates the subpopulations, and that a Neyman–Pearson gate composed with the map is jointly optimal in its class. A sequential form of the gate decides from a prefix, so the operator runs in a single pass. On reasoning confidence in gemma-2-2b-it the additive edit removes every overconfident-wrong answer only by destroying the confident-right ones, which the token-level Cohen’s d of 0.049 we measure between the two makes a geometric necessity rather than a tuning failure. Across five resampled MMLU subsets our conditional operator reaches selectivity $+0.279$ at no accuracy cost, transfers to ARC-Challenge and GSM8K without re-fitting, and replicates at $4.5\times$ scale. Giving every baseline an equal tuning budget, we find it keeps that selectivity when the questions change, where a tuned full-space map loses two thirds of its own. It also leaves capability intact where the additive edit does not, staying at the unsteered value on HumanEval, WikiText-103 and open-ended generation, and a held-out OLMo-2/TruthfulQA evaluation supports the detector axis beyond Gemma.

§1. Introduction

Inference-time activation steering promises behavior control without retraining: add a direction to the residual stream and the model refuses less, hedges more, or changes style. The workhorse is contrastive activation addition (CAA) [20]: a fixed vector, a hand-tuned coefficient, applied to

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every token. Yet applied to a concrete reasoning behavior — overconfidence of gemma-2-2b-it on multiple-choice reasoning — CAA fails in a way no coefficient tuning fixes: the dose that eliminates overconfident-wrong answers (OCW $50 \rightarrow 0$ under a calibrated readout) simultaneously eliminates confident-right ones (CR $76 \rightarrow 1$) and costs 8.7 accuracy points.

The failure is not that additive steering is a weak tool. On refusal it is a strong one, and prior work models that behavior as a single-direction shift [1, 20], and a shift is exactly what an additive edit implements optimally. Overconfidence asks for something else. The target is not a relocated distribution but a compressed one: the tail must move while the bulk stays. Corollary 1 covers both cases with one statement, and reading it in the right direction is what separates the successes from this collapse.

This paper treats that failure as a mathematical object rather than an engineering nuisance. Our contributions are as follows. First, we give a unifying framework with optimality theorems. Every linear steering operator is a scalar transfer function g on the projection $p = \langle h, \hat{v} \rangle$, with additive steering $g(p) = p + \alpha$, ablation $g = 0$, and CAST-style gating [13] $g(p) = p + \alpha \mathbf{1}[p > \tau]$. We prove in Thm. 1 that the optimal-transport map between projection laws is the unique minimal-perturbation intervention, and in Cor. 1 that additive steering is optimal only for pure location shifts. Second, we give a quantified impossibility result and its repair. Thm. 2b caps the selectivity of any transfer function on an axis by the separability of the subpopulations along it, and we measure a token-level Cohen’s $d = 0.049$ between OCW and CR on the confidence axis, so the additive collapse is a geometric necessity rather than a tuning failure. The repair is conditional. An LDA gate on a second, detection axis is Neyman–Pearson optimal, and gate $\times$ transport is jointly optimal in its class (Thm. 3), while Thm. 4 ties an online decision to the post-hoc one at a stated rate and so keeps the whole pipeline single-pass. Third, we obtain strong trade-offs at minimal action cost in the evaluated setting. Across five MMLU seeds our conditional operators reach selectivity $+0.22$ – 0.28 at zero accuracy cost, against $+0.00$ for additive steering at its published dose and $+0.20$ for prompting (Sec. 5). Given every method an equal tuning budget they keep that selectivity when the questions change, where a faithful reimplement of MiMiC keeps half of its own, and they retain more CR than any baseline that matches them (Sec. 5). They replicate at $4.5\times$ scale, transfer to ARC-Challenge and GSM8K with no refitting (Sec. 6), and every operator is $O(d)$ per token with 19 KB of state, the optimal order by Prop. 3. Our results are a first

account of why a steering edit damages what it was not aimed at, and of how much of that damage the geometry forces rather than the tuning.

§2. Related Work

Linear steering. ActAdd [29] and CAA [20] add fixed diff-in-means vectors, ITI [14] shifts probe-selected heads, RepE [35] generalizes reading and control vectors, directional ablation removes one direction everywhere [1], and SAE clamping fixes a feature value [28]. All are special cases of our transfer-function family, and all are input-independent edits. AxBench [33] finds them weaker than prompting on instruction-following, which is the concrete gap this family has to close. CAST [13] adds conditional gating, but its action remains additive, and SADI [32] and KL-adaptive composition [23] adapt dose per input without changing its shape. In this paper we take the shape of the edit as the object of study, and derive locality, dose control and serving cost from one optimality statement rather than fixing them by hand.

Distribution matching and optimality. MiMiC [24] derives the Gaussian- W_2 affine map, applied unconditionally in the full space, AcT [21] fits per-neuron 1-D transport, and LinEAS [22] learns affine maps under a distributional loss. COAST [19] proves pointwise optimality of a covariance-weighted projection, which in our language is transport to a point mass under the M of Prop. 2, and A-LQR [18] proves LQR-optimality of a control input. We prove a different statement in Theorem 1, namely the optimal map between distributions under a locality constraint, together with the conditional optimality and selectivity ceiling of Thms. 2–3 that no prior steering work states. PCHI [15] trains probe-conditioned head rescalings for the same behavior, and its additive baseline corrects $\approx 0\%$, independently confirming Cor. 1. That baseline is ungated, so its design leaves the gated-additive cell empty and confounds gating with the multiplicative head parameterisation, and our grid fills exactly that cell. It also contains a probe-conditioned member of the same family, a full- d logistic gate composed with each action (Appendix J). [34] reports the dose-response our per-sample dose exploits.

Deciding during generation. We are not the first to move the decision inside the loop. CAST [13] compares each token’s hidden state with a condition vector at a tuned threshold, DSAS [4] computes context-dependent per-token scales and states the same when-versus-how split we use, and GAPS [6] adds dimension-level gates at $O(d)$ per token. A prediction probe

read after a few generated tokens can anticipate whether steering will succeed [25], and YOPO [16] removes a second clean pass by reconstructing the unsteered residual before reading its direction. What none of these provides is a decision with an error guarantee: our gate is a Neyman–Pearson test whose online decision is tied to the post-hoc one at a stated rate (Thm. 4). Two findings in this literature shape our design. Reading a direction off already-steered activations degrades it [16], so our reader sits above the action and the decision is a single commitment, after which the reader’s output is no longer used. And a feature that detects a behavior in finished text is a poor predictor of it from a prefix [5], so we do not reuse the whole-trace head online: we refit it on the prefix budget the gate will actually have, and report the budget–AUROC curve (Appendix G).

§3. Method: Projection-Transport Steering

We write $h \in \mathbb{R}^d$ for the residual-stream activation at a fixed decoder layer and let $V \in \mathbb{R}^{k \times d}$ have orthonormal rows spanning a behavioral subspace. The two axes that carry the behavior are a confidence axis and a correctness axis, obtained by diff-in-means on the model’s own labeled reasoning traces; k itself is a design parameter that we select rather than fix. We write $q = Vh \in \mathbb{R}^k$, let μ be the law of h under natural generation, μ_V the induced law of q , and ν a target law on $\mathbb{R}^k$.

Definition 1 (admissible steering). $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is admissible for (V, ν) if (i) $(I - V^\top V)T(h) = (I - V^\top V)h$ μ -a.s. (locality: the orthogonal complement is untouched), and (ii) $(V \circ T)_\# \mu = \nu$, i.e. the pushforward of μ equals the target law (goal: steered projections follow the target law). The cost is $C(T) = \mathbb{E}_\mu \|T(h) - h\|^2$.

We regularize by that cost because large interventions are what empirically destroy generation, as we show in Sec. 5, and because under a quadratic functional metric C bounds the KL drift of the next-token distribution (Prop. 2). The first result says that once the subspace and the target law are fixed, there is nothing left to design.

Theorem 1 (steering is optimal transport of the projection). $\min_T C(T) = W_2^2(\mu_V, \nu)$ over admissible T , attained by $T^*(h) = h + V^\top(S(Vh) - Vh)$ where S is the (μ_V -a.s. unique) optimal Monge map from μ_V to ν . Any optimal T satisfies $VT(h) = S(Vh)$ almost surely, so the optimal intervention is a deterministic function of the projection alone and conditioning on $h_\perp$ cannot reduce the cost.

Proof sketch. Locality collapses the cost to the projection, and $(Vh, VT(h))$ couples (μ_V, ν) , so $C(T) \geq W_2^2$. Brenier’s map attains it. Full proof in Appendix B. $\square$

For $k = 1$ the map $S = F_\nu^{-1} \circ F_{\mu_V}$ is the monotone quantile map, and we implement it as a 41-knot piecewise-linear lookup. For Gaussian marginals it is affine [7, 12], $S(q) = m_t + A(q - m_s)$ with $A = \Sigma_s^{-1/2} (\Sigma_s^{1/2} \Sigma_t \Sigma_s^{1/2})^{1/2} \Sigma_s^{-1/2}$.

Choosing the subspace. Theorem 1 holds for any orthonormal V , so k is a design parameter and not a fixed property of the behavior. Two interpretable axes make the geometry legible and are what the gate needs, but the action is free to use more. The natural larger choice is the subspace an unconditional full-space map actually moves: the mean shift $m_t - m_s$ together with the leading eigenvectors of $A - I$, where A is the Bures–Wasserstein matrix of the source and target moments. Retaining the interpretable axes and then the first k of those directions costs $\Theta(kd)$ per token instead of $\Theta(d^2)$. At the far end $V = I$ recovers the full-space Gaussian map that MiMiC [24] applies unconditionally, so one family runs from a single interpretable axis to a dense $\Theta(d^2)$ operator and Theorem 3 gates every point of it. We select k on the tuning split like every other hyperparameter, and Sec. 5 reports what that buys, which on this behavior is less than the construction suggests (Appendix H).

Reading Theorem 1 backwards tells us when the workhorse edit is already the right one, and when it cannot be.

Corollary 1 (additive steering is forced to be global). Additive steering $g(p) = p + \alpha$ is the optimal map iff $\nu = \mu_V * \delta_\alpha$ (pure shift). If the target compresses the spread, which is what “hedge only abnormal confidence” requires ($\sigma_t < \sigma_s$), the optimal slope is $\sigma_t/\sigma_s < 1$: the tail must move more than the bulk. A slope-1 map matched to the tail over-transportes the bulk by the same constant, which is the observed CR collapse. Moreover its selectivity between two subpopulations with equal projection laws is identically zero.

If no function of the confidence coordinate can separate the two subpopulations, we need a second coordinate and a decision rule on it. The next result says what the best such rule is, and how much it can buy.

Theorem 2 (optimal gate and selectivity ceiling). Let a gate fire iff a trace statistic $s > \tau$, with $\text{TPR}(\tau) = P(s > \tau \mid \text{OCW})$, $\text{FPR}(\tau) = P(s >$

$\tau \mid CR$). (a) For any objective increasing in TPR and decreasing in FPR, the likelihood-ratio test is optimal at every operating point (Neyman–Pearson). Under equal-covariance Gaussian class conditionals it reduces to the linear discriminant $w = \Sigma^{-1}(\mu_{OCW} - \mu_{-OCW})$. (b) For any statistic with concavified ROC area AUC, $\max_{\tau}[\text{TPR} - \text{FPR}] \leq 2 \text{AUC} - 1$.

Gate and action can then be designed separately without losing optimality.

Theorem 3 (joint optimality of gate $\times$ transport). Among interventions that are local to V , identity off the gate event $\Gamma = \{s > \tau\}$, and achieve target law ν_{Γ} on Γ , the cost-minimal intervention applies Theorem 1’s map for the conditional law $(\mu_V \mid \Gamma) \rightarrow \nu_{\Gamma}$ on Γ ; its cost is $P(\Gamma) W_2^2(\mu_V \mid \Gamma, \nu_{\Gamma})$.

Theorem 3 treats the gate as a switch. Nothing forces that: the same cost functional, asked for the best dose rather than the best event, answers with a graded rule whose threshold is measured instead of tuned.

Proposition 1 (the optimal dose is affine in the gate posterior). Let $T_{\lambda}(h) = h + \lambda V^{\top}(S(Vh) - Vh)$ interpolate between the identity and the map of Thm. 1, and let the dose λ be any $[0, 1]$ -valued function of the gate statistic s . Write $\pi(s) = P(OCW \mid s)$, $d(s) = \mathbb{E}[\|S(Vh) - Vh\|^2 \mid s]$, and let the behavioral gain of a dose be linear in it with class slopes β_1 on OCW and $-\beta_0$ on CR. Then the dose maximising gain minus displacement priced at $1/2c$ is

$$\lambda^*(s) = \left[\frac{c(\beta_0 + \beta_1)}{d(s)} (\pi(s) - \tau) \right]_0^1, \quad \tau = \frac{\beta_0}{\beta_0 + \beta_1},$$

nondecreasing in π , zero exactly on $\{\pi \leq \tau\}$, and converging pointwise to $\mathbf{1}[\pi(s) > \tau]$ as $c/d \rightarrow \infty$. The hard gate is thus the free-displacement limit of the graded operator, and it is strictly suboptimal at any finite price of displacement whenever π has no atom at τ .

Proof sketch. The dose is s -measurable, so the objective splits as an integral over s of a strictly concave quadratic in λ ; maximise it pointwise on $[0, 1]$. Full proof in Appendix B. $\square$

Two things follow that the switch does not give us. The operating point τ is a ratio of two measured rates, the collateral and the removal a full dose produces, so it is read off the tuning split rather than searched; and because the linear discriminant of Thm. 2a is the log-likelihood ratio, $\pi(s)$ is $\sigma(s)$ up to the prior term, which we fit on the extraction traces. One

slope $\rho = c(\beta_0 + \beta_1)/d$ is left to search, in place of a threshold and an action.

Theorem 3 says nothing about when Γ must be decided. Scoring a finished trace and regenerating it is one implementation. Deciding while the trace is written is another, and it is the one a server can afford. Write z_t for the mean projection over the first t generated positions and z_∞ for the completed-trace summary the post-hoc gate uses.

Theorem 4 (single-pass gating). (a) Among all gates measurable with respect to the first t positions, the likelihood-ratio test on z_t is optimal at every operating point, and under equal-covariance Gaussian class conditionals it is the linear discriminant fitted on z_t — Theorem 2 applied to the statistic the gate actually has. Its ceiling is $2 \text{AUC}(z_t) - 1$, which is what a prefix budget costs. (b) If instead the gate insists on the whole-trace statistic, let the contributions form a martingale difference sequence around z_∞ with conditionally σ -sub-Gaussian increments and fire at the first t with $s_t > \tau + B_t(\delta)$, where $B_t(\delta) = 1.7 \sigma \sqrt{(\log \log 2t + 0.72 \log(5.2/\delta))/t}$ is the stitched finite-law-of-the-iterated-logarithm boundary of Howard et al. [11]. Then it fires on a trace with $s_\infty \leq \tau$ with probability at most δ , simultaneously over all t , and on a trace with margin $\Delta = s_\infty - \tau > 0$ it fires by the first t with $B_t(\delta) \leq \Delta/2$, after $O(\sigma^2 \Delta^{-2} \log \log(1/\delta))$ positions. (c) Either way the composed operator is single-pass: the action starts at the firing position, costs $O(d)$ per token, keeps $O(1)$ extra state, and invalidates no cache entry, because positions written before the switch keep the keys they were written with.

Proof sketch. Part (a) is Theorem 2 with z_t in place of z , part (b) is the time-uniform confidence sequence of Howard et al. [10, 11], and part (c) follows from the order of the two hooks. We give the full proof in Appendix B. $\square$

Remark 1 (detection and action need not share a depth). Nothing in Theorems 1–4 forces the gate to read the layer the action writes. Placing the reader above the actor costs one position of lag and lets each sit where it is best. On gemma-2-2b-it the detection contrast peaks at layer 16, where the trace-level AUROC is 0.799 against 0.769 at layer 14, and the gap is larger for the prefix heads the sequential gate uses (0.740 against 0.712 at 16 generated tokens), so we let the reader sit at layer 16 while the action stays at layer 14 (Appendix G). The order also removes an interference

the single-pass setting would otherwise have: reading a direction off activations the same operator has already edited degrades it [16], and here the decision is a single commitment taken before anything is edited.

Two properties of the operator remain to be stated: that it does not depend on the Euclidean metric, and that it is as cheap as any such operator can be.

Proposition 2 (metric covariance). If cost is $(\Delta h)^\top M(\Delta h)$ with $M \succ 0$ (e.g. a Fisher/Gauss–Newton pullback, so cost approximates the induced next-token KL), the admissible problem reduces to optimal transport of μ_V to ν inside $\mathbb{R}^k$ under the Mahalanobis cost $M_V = VMV^\top \succ 0$: the optimal map is $S_M = M_V^{-1/2} \circ \nabla \varphi \circ M_V^{1/2}$ with φ convex, and the operator is again $T^*(h) = h + V^\top(S_M(Vh) - Vh)$. Euclidean PTS is $M_V = I_k$. Proof in Appendix B.

Remark 2 (support preservation). The quantile map never leaves the target’s empirical support. An additive map violates support for every $\alpha \neq 0$, consistent with the observed generation collapse at $|\alpha| = 1.5\|h\|$.

Proposition 3 (computational optimality). Any algorithm implementing a non-degenerate intervention local to a subspace spanned by a dense direction must read all d coordinates of h , i.e. costs $\Omega(d)$ per token. PTS costs $\Theta(kd + \log K)$, which matches that lower bound for constant k and K , against $\Theta(d^2)$ for a full-space affine map (MiMiC), $\Theta(dd_{\text{sae}})$ for SAE steering, and a full extra forward pass for KL-adaptive dosing. Proof in Appendix B.

§4. Experimental Setup

Models, datasets, and splits. The main experiments use gemma-2-2b-it [8], and the central comparison is repeated on gemma-2-9b-it at 4.5× scale (Sec. 6).¹ Both models reason inside `<think>...</think>` and answer `\boxed{letter}`. Decoding is greedy, so evaluation “seeds” resample questions, not decoding noise. Activations are read and modified at layer 14/26 (2B) and 24/42 (9B). Benchmarks: MMLU [9] (fitting and primary evaluation), ARC-Challenge [2], and GSM8K [3] as 4-option MCQ with seeded distractors (GSM8K-MCQ, Appendix C), $n=300$ questions each, disjoint from all fitting data. We fit directions, maps, and gate parameters

¹2B weights were served from the unsloth/gemma-2-2b-it mirror. We verified all 288 tensors are bit-identical to the official google/gemma-2-2b-it release.

once per scale, on a 400-question MMLU extraction split ($\sim 50k$ trace tokens), and reuse them everywhere, so cross-benchmark results are transfer rather than refits. We run the main comparison on five resampled MMLU subsets and the 9B replication on three.

Behavioral states and confidence readouts. Every answer gets one of four labels — overconfident-wrong (OCW), nonconfident-wrong, nonconfident-right, confident-right (CR) — under two independent readouts: the letter-logit readout M2 (softmax of the letter logits at the trace’s own answer position) and the yes-probability readout M4 (normalized $P(\text{YES})$ over per-option yes/no reformulations: "Answer whether the given answer is the right answer. . . Question: {q} Answer: {option}", one forward per option, reasoning in $\langle \text{think} \rangle$, then YES/NO, and the four $P(\text{YES})$ values are renormalised). M3 is the model’s own stated probability, elicited after the trace with "How confident are you in this answer? Reply with a probability between 0 and 1." On identical questions they disagree sharply (ECE 0.39 against 0.10, Appendix D): M2 is strongly overconfident, M4 much better calibrated. M4 is therefore the primary readout, and agreement under M2 and the M3 self-report is a robustness requirement. Full grids: Appendix J.

Directions and the trace-level gate. On the extraction split we fit a confidence axis $\hat{v}$, the steering axis, and a detection axis $\hat{u}$, both by diff-in-means, and we orthonormalize them into $V = (\hat{v}, \hat{u})$. We keep the two contrasts disjoint by construction. The first separates the two confident states $\{OCW, CR\}$ from the two nonconfident ones, so $\hat{v}$ encodes confidence irrespective of correctness, and the second separates OCW from CR, so $\hat{u}$ encodes whether that confidence is warranted. Layer 14 is the depth at which the second contrast is most linearly readable (Appendix G). A completed trace of length T is summarized by its mean projections $z = (\frac{1}{T} \sum_t \langle h_t, \hat{v} \rangle, \frac{1}{T} \sum_t \langle h_t, \hat{u} \rangle)$, and an LDA head gives the gate score $s(z) = w^\top z + b$, firing iff $s(z) > \tau$. Here w and b come from the equal-covariance LDA solution and τ is a quantile of the extraction-split score distribution (the median unless stated), never a quantity fitted on evaluation data. We run the gate in two regimes. Post-hoc (two-pass): generate unsteered, keep that output if the gate is silent, and otherwise regenerate with the intervention active during reasoning. Sequential (single-pass, Thm. 4): a second LDA head, fitted on the mean projection over the first t^* generated positions rather than over the finished trace, is read at layer 16 while the action runs at layer 14 (Remark 1), and the action switches on at token t^* and stays on. Fitting the head on the statistic the

gate will actually have is what makes an early decision sound, and t^* and the threshold quantile are searched on the tuning split like every other hyperparameter. Nothing is regenerated. In both regimes the gate sees only the model’s own activations, with no gold labels at inference.

Steering methods. All operators act through $p_t = \langle h_t, \hat{v} \rangle$ (Sec. 3). Additive is $h'_t = h_t + \alpha \hat{v}$ at $\alpha = -0.75\|h\|$, the dose that eliminates OCW under the calibrated readout. Ablation is $h'_t = h_t - p_t \hat{v}$, and clamp is $p'_t = \min(p_t, c)$ at a source quantile, with the median as the representative condition. Quantile, Gaussian and Bures–Wasserstein transport map onto a target law fitted on the extraction split. We form conditional variants by composing the trace gate with each action, giving CAST-style gate+additive, gated clamp, gated ablation, gated OT and LDA-gated subspace BW. In each of them the gate controls exposure and the action controls severity, especially on false positives. We compare against prompting, additive CAA [20], ablation [1], CAST-style [13], MiMiC [24] and Linear-AcT per-neuron transport [21], all reimplemented and run on identical records, decoding, readouts, and metrics. Every method, ours included, is given the same tuning budget of eight configurations, searched on the same held-out MMLU tuning split under one pre-declared rule. CAST gets its threshold and dose, MiMiC and Linear-AcT get layer selection, and our gate gets the depth it reads and the quantile it fires at (Appendix H). We also run think-only variants, which disable steering before the answer tokens.

Metrics and statistical protocol. We choose each metric to answer one question about the intervention.

Accuracy asks whether it hurt task performance, and we report the change Δacc against the unsteered model.

ECE, the expected calibration error, asks whether the confidence numbers can be trusted. We compute the mean absolute gap between average stated confidence and actual accuracy over 10 confidence bins, where lower is better and 0 means confidence matches reality.

The state histogram asks where the answers moved, and we report the counts of the four behavioral states before and after intervention.

The surgical metrics ask whether it hit only its target. Pairing answers by question, $\text{ocw_rm} = P(\text{left OCW} \mid \text{was OCW})$ is the fraction of the harmful state removed, $\text{cr_keep} = P(\text{stayed CR} \mid \text{was CR})$ the fraction of the healthy state preserved, and selectivity $\text{Sel} = \text{ocw_rm} - (1 - \text{cr_keep})$ combines them (+1 is a perfect scalpel, 0 is indiscriminate).

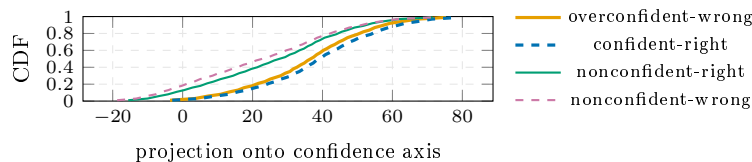


Figure 1. Token-level projection CDFs on the confidence axis by behavioral state (extraction split, $\sim 50k$ tokens). The OCW and CR curves are nearly identical (Cohen’s $d = 0.049$), while the confident and nonconfident pairs are well separated ($d \approx 0.6$).

Selective prediction, which we measure by AURC and E-AURC, asks whether confidence still ranks right answers above wrong ones. AURC is the area under the risk–coverage curve, where lower is better. Monotone post-hoc recalibration leaves it invariant, so only interventions that change answers or their ranking can move it.

Single-subset results carry paired-bootstrap 95% CIs, and multi-subset results are mean $\pm$ std over independently resampled subsets.

Post-review validation protocol. We froze a simplified detector-axis action after a development-only Qwen3 component analysis, and we then evaluated it on held-out TruthfulQA panels for Mistral-7B, OLMo-2-7B, and Granite-8B. Each eligible architecture used the same 237-question composition (50 in-domain, 187 shifted) and was compared with a norm-matched spherical action using 10,000 paired BCa resamples, and the three architecture comparisons used Bonferroni-adjusted two-sided intervals. OLMo-2 subsequently faced seven pre-frozen construction and specificity controls. A separate six-cell Mistral/OLMo-2/Granite $\times$ BBQ/ETHICS observer screen kept fitting, screening, and confirmatory units disjoint, and only Mistral/ETHICS advanced to a 1,000-scenario confirmatory test, with intervals adjusted for the six planned cells. Only observer-eligible cells were opened, and the planned denominator remained six.

§5. Results on MMLU

The confidence axis does not separate justified from unjustified confidence. We first test the geometric premise of selective steering. In Figure 1 the OCW and CR projection CDFs nearly coincide at $d = 0.049$, while the confident and nonconfident pairs separate at $d \approx 0.6$. We read this as the axis encoding whether the model is confident and not whether

that confidence is warranted, so no transfer function of this coordinate can be surgical (Cor. 1 and Thm. 2b at $AUC \approx .51$). The measurements agree: the selected dose removes all 50 OCW but takes CR from 76 to 1. Across five subsets additive selectivity is $+0.000 \pm 0.000$, accuracy drops 8.7 ± 5.3 points, and AURC worsens from 0.404 to 0.556. We therefore do not read the removal of OCW as evidence of calibration, because the shift suppresses desirable and undesirable confidence alike. The reliability literature reports the same quantity from the other side: steerability varies by behavior rather than by prompt construction, and what predicts it is how well the two subpopulations separate along the difference-in-means line [26, 27]. Thm. 2b is the statement that this is not a correlation to be improved by better fitting but a ceiling.

Projection-dependent actions reduce collateral damage. Median clamping — one scalar touching only the abnormal tail — reaches selectivity $+0.124$ on the representative subset at $+1.3$ accuracy points. Across five subsets accuracy is $+0.001 \pm 0.034$, that is preserved rather than improved (single-split accuracy gains should be treated as noise until seed-replicated), while selectivity holds at $+0.145 \pm 0.039$. Quantile transport to the non-OCW target reaches $+0.119 \pm 0.086$, likewise without the additive loss. We read the lesson as one about shape rather than strength. Abnormal high-confidence activations must move more than ordinary ones, and ablation, clamp, and transport achieve that by conditioning on the input projection.

Main result. Across five MMLU subsets we find that gated ablation attains selectivity $+0.279 \pm 0.090$ (additive $+0.000 \pm 0.000$, prompting $+0.198 \pm 0.086$) at no detectable accuracy cost (-0.003 ± 0.050), with the best ECE ($.094 \pm .030$) and AURC ($.395 \pm .038$), beating faithful reimplementations of CAST-style, MiMiC, and Linear-AcT at their published configurations on every column (Table 1, Fig. 2).

We find tuning changes what the baselines are. The additive dose we select is a third of the published one and the CAST dose is half, and at those doses both stop being destructive: additive reaches $+0.256 \pm 0.133$ and CAST $+0.227 \pm 0.082$ at no accuracy cost. We read that as the useful result of running the protocol rather than a threat to the framework, because what the two buy their selectivity with is visible in the same row. Additive retains 0.621 of CR and CAST removes only 0.432 of the OCW it fires on, the two ways Cor. 1 says a shift can trade. MiMiC holds the residual stream together but keeps half its tuning-split selectivity once the

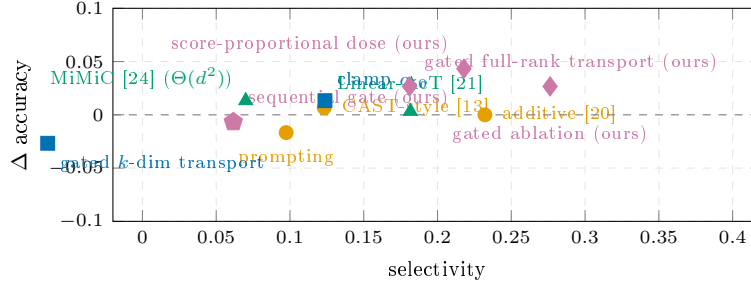


Figure 2. The selectivity-accuracy frontier under the identical MMLU protocol (M4, representative subset, with multi-subset means in Tab. 1). Additive-action methods (orange) pay accuracy for selectivity or achieve neither, ungated distribution matching (green) is safe but weakly selective, and gate $\times$ tail-local action (ours) sits above both at $\Theta(kd)$ per token against $\Theta(d^2)$ for a full-space map (Prop. 3; at this scale that difference is below the wall-clock noise floor, Appendix M).

method (M4, MMLU, seeds mean $\pm$ std)	Δ acc	ECE	AURC	cr_keep	Sel
baseline	acc 0.473 $\pm$ 0.026	0.099 $\pm$ 0.019	0.404 $\pm$ 0.029	—	—
prompting	-0.005 $\pm$ 0.020	0.118 $\pm$ 0.018	0.392 $\pm$ 0.023	0.735 $\pm$ 0.063	+0.198 $\pm$ 0.086
additive CAA [20]	-0.004 $\pm$ 0.041	0.080 $\pm$ 0.017	0.400 $\pm$ 0.038	0.621 $\pm$ 0.067	+0.256 $\pm$ 0.133
CAST-style [13]	-0.007 $\pm$ 0.032	0.072 $\pm$ 0.012	0.396 $\pm$ 0.017	0.795 $\pm$ 0.031	+0.227 $\pm$ 0.082
MiMiC [24]	-0.015 $\pm$ 0.045	0.113 $\pm$ 0.010	0.408 $\pm$ 0.040	0.763 $\pm$ 0.031	+0.132 $\pm$ 0.103
Linear-AcT [21]	+0.001 $\pm$ 0.032	0.101 $\pm$ 0.011	0.410 $\pm$ 0.027	0.868 $\pm$ 0.048	+0.146 $\pm$ 0.141
gated local action (ours)	-0.015 $\pm$ 0.036	0.089 $\pm$ 0.013	0.413 $\pm$ 0.037	0.713 $\pm$ 0.050	+0.206 $\pm$ 0.059
gated full-rank transport (ours)	+0.003 $\pm$ 0.021	0.108 $\pm$ 0.022	0.394 $\pm$ 0.044	0.854 $\pm$ 0.038	+0.146 $\pm$ 0.042
score-proportional dose (ours)	-0.005 $\pm$ 0.054	0.089 $\pm$ 0.010	0.406 $\pm$ 0.050	0.650 $\pm$ 0.056	+0.231 $\pm$ 0.120
sequential gate (ours, single pass)	+0.009 $\pm$ 0.015	0.111 $\pm$ 0.018	0.398 $\pm$ 0.024	0.897 $\pm$ 0.017	+0.079 $\pm$ 0.050

Table 1. Multi-seed head-to-head at equal tuning budget: every method runs the configuration it won on the held-out tuning split (Appendix H), over five evaluation subsets of $n=300$. Greedy decoding is deterministic, so seeds resample questions, a stricter robustness axis than sampling seeds. Only Linear-AcT reports seed variance in this literature. The default-configuration grid is Table 11.

questions change (+0.132 $\pm$ 0.103) and rewrites the stream at $\Theta(d^2)$ per token, 1152 $\times$ the cost of a $k=2$ edit. Prompting reaches +0.198 $\pm$ 0.086 and Linear-AcT +0.146 $\pm$ 0.141. Our gated local action holds +0.206 $\pm$ 0.059 with retention 0.713 and the smallest spread in the table, and the score-proportional dose reaches +0.231 $\pm$ 0.120. We do not claim per-metric maxima in every run; we claim the strongest joint trade-off across removal,

retention, accuracy and selective prediction (Fig. 2), and the only operators here that cost nothing at all where the behavior is absent (Table 3).

The comparison survives equal tuning. The first question to ask of a head-to-head is whether the baselines were tuned as hard as the method, so we gave every operator the same budget of eight configurations on the same held-out tuning split under the same selection rule (Appendix H). Tuning moves the baselines, and in one case a long way. The additive dose falls from $-0.75\|h\|$ to $-0.25\|h\|$ and its tuning-split selectivity rises from ≈ 0 to $+0.271$, which is worth stating plainly: the additive edit is not useless, it was over-dosed, and the dose that removes every OCW is far past the dose that trades best. MiMiC reaches $+0.263$ at layer 18, CAST $+0.184$ at the CR 70th percentile and Linear-AcT $+0.184$ at layer 14. Our gated ablation fires at the 30th rather than the 50th percentile for $+0.245$, and the score-proportional dose of the next paragraph reaches $+0.267$, within noise of the tuned additive dose and above every other method. Table 1 carries those configurations, unchanged, to the five evaluation subsets.

The subspace dimension is not the lever. Theorem 3 says our gate composes with any transport, so a full-space map is not a rival family but the $k=d$ member of ours, and we tuned the family end to end under one budget. We measure selectivity peaking at $k=2$, $+0.186$, and flat above it, $+0.148$ at $k=8$ and again at $k=32$, so the two interpretable axes already carry this behavior and the extra 2302 coordinates buy nothing. The $k=d$ member says the same thing from the other end. We measure the gate composed with MiMiC’s own map at $+0.286$ on a 400-question split and $+0.155$ on the 1200-question one, the largest movement of any configuration we searched (Table 9), which is what a configuration selected on noise looks like when the noise is removed. We keep $k=2$ because the data says the behavior lives there, not because the operator is cheaper there. The other members of the family, a token-local action and a posterior-weighted dose, are tuned and carried the same way in Appendix I.

What survives a change of questions. Carried unchanged to five re-sampled question sets, the tuning-split ordering does not hold. MiMiC keeps half of what the layer search bought it, $+0.263$ on the split and $+0.132 \pm 0.103$ on the subsets, and gating the same map stabilises it at a lower level, $+0.155$ and $+0.146 \pm 0.042$. Our gated local action moves from $+0.245$ to $+0.206 \pm 0.059$ and the score-proportional dose from $+0.267$ to $+0.231 \pm 0.120$. A $\Theta(d^2)$ map fitted at one layer is reading the sample, and a two-dimensional edit behind a gate has little to overfit.

Spending the edit where the evidence is. What the tuning data points at is retention rather than removal: at its selected threshold the hard gate fires on most flagged traces and full ablation on the marginal ones is what costs CR. So we let the strength of the action follow the gate score instead of switching on it, the interpolation g_λ the method already defines, and that is the best configuration any method reaches under the budget, +0.267 against +0.271 for the tuned additive dose and +0.263 for MiMiC. It does not win the way the diagnosis suggested. The selected configuration widens exposure instead of narrowing it, firing at the 10th percentile and scaling each edit by its margin, so grading the dose is what makes a permissive threshold affordable. On the evaluation subsets it reaches $+0.231 \pm 0.120$, and on ARC-Challenge, off the split it was tuned on, it is the best member we have at +0.342 with retention 0.815, against +0.290 for the hard switch.

Gate and action shape are complementary. We can see why gate and local action compose in the measured movement of real regenerated traces (Fig. 3, Appendix E). Additive steering moves all 50 OCW far past the calibrated zone and drags all 76 CR along. Gated ablation protects twice: 30 CR below threshold stay bitwise identical (77/300 questions overall), and of the 46 CR it wrongly flags, 25 keep their state because the tail-local action barely moves typical activations. CAST-style has the first protection but not the second, and ungated clamp or transport has the second but not the first.

Detection is the main remaining bottleneck. We find the LDA gate separates OCW from CR at trace-level AUROC ≈ 0.77 : it catches 41/50, misses 9, and false-fires on 46/76 CR. The diagnostic value $2 \text{ AUROC} - 1 \approx 0.538$ is close to twice the achieved selectivity. Thm. 2b applies to the concavified ROC area, so raw AUROC is not an exact behavioral ceiling. Detection, rather than the transport map, is nevertheless the binding empirical constraint, and Fig. 4 (bottom) plots the gap against the label oracle. Five different actions inside the same gate buy removal at the same rate in retention (Appendix I), which is the practical form of that constraint. Because the constraint is detection, we ask where detection is best. We refit the gate at nine depths (Appendix G) and find the separation is not monotone in depth. It peaks at layer 16, where AUROC is 0.799 and the ceiling 0.599, and collapses by layer 25 to 0.649, where the residual stream has already committed to an answer. Since the framework never required the gate to read the layer the action writes, we let it read layer 16 and keep acting at layer 14.

The edit costs nothing outside the behavior it was fitted for. Selectivity inside MMLU says nothing about what an always-on edit does to generation it was never meant to touch, so we measure three tasks the directions never saw (Appendix K). We find that the dose that removes every OCW takes HumanEval pass@1 from .396 to .128, doubles WikiText-103 perplexity ($23.4 \rightarrow 47.3$), and makes 57.5% of 200 open-ended generations degenerate (distinct-3 $.953 \rightarrow .480$). A judge asked to compare the steered and unsteered answers to the same instruction, in both presentation orders, prefers the unsteered one on 185 of 200 prompts. We find the projection-local actions invisible on all four measurements even when left on for every token, at pass@1 .390–.396, perplexity 23.0–23.4, distinct-3 .946–.951, order-consistent win rates of .470–.508 against a tie value of .5, and no degenerate generation. MiMiC is close behind at pass@1 .396 and win rate .512, for $\Theta(d^2)$ per token. The equal-budget dose is the fair comparison and it tells the rest of the story: at $-0.25\|h\|$ the additive edit is no longer destructive, at pass@1 .408, perplexity 23.6, win rate .495 and no degenerate generation, so what the parity protocol selects is a dose too small to break the model. We still find it the wrong shape. That dose leaves 69.9% of MMLU tokens outside the 99% conditional ellipsoid against 0.9% unsteered and 1.0–1.3% for every projection-local operator (Appendix L), and it cannot abstain: our sequential gate fires on none of HumanEval and none of the teacher-forced WikiText, so in those two rows we report the unsteered numbers to the digit. Because the target law leaves the bulk in place, we expect a projection-local operator to have nothing to do where the behavior is absent, and that is what we observe.

Robustness across instruments, and think-only steering. Absolute counts vary across readouts, but the conclusions persist: gated ablation reaches selectivity +0.35 (M4), +0.15 (M2), +0.17 (M3 self-report: 37% of OCW removed at 0.80 retention). Think-only variants keep every effect’s direction at near-zero accuracy cost: the intervention acts through the reasoning trajectory, not the answer logits.

§6. Scale, Transfer, and Serving Cost

Replication at 9B. We find the 2B story replicates on gemma-2-9b-it (Table 2): additive steering has identically zero selectivity and CR-retention on every subset while losing 18.4 ± 3.3 points. CAST-style buys $+0.283$ selectivity for 11.4 ± 3.2 points and 0.45 retention, while the local operators alone gain accuracy and improve selective prediction (AURC

method (M4, 9B, 3 seeds)	Δacc	ECE	cr_keep	AURC	Sel
baseline	$\text{acc } 0.633 \pm 0.031$	0.144 ± 0.021	—	0.197 ± 0.025	—
additive -0.75	-0.184 ± 0.033	0.167 ± 0.037	0.000 ± 0.000	0.450 ± 0.023	$+0.000 \pm 0.000$
CAST-style gate+add	-0.114 ± 0.032	0.101 ± 0.011	0.453 ± 0.038	0.285 ± 0.016	$+0.283 \pm 0.093$
clamp q_{50} (ours)	$+0.023 \pm 0.012$	0.143 ± 0.018	0.887 ± 0.015	0.184 ± 0.025	$+0.202 \pm 0.020$
gated ablation (ours)	$+0.011 \pm 0.008$	0.127 ± 0.009	0.907 ± 0.014	0.186 ± 0.023	$+0.203 \pm 0.034$

Table 2. gemma-2-9b-it, MMLU, layer 24, $n=300 \times 3$ seeds:
the head-to-head replicates at $4.5\times$ scale.

0.184–0.186 against 0.197), at selectivity +0.20 and retention 0.89–0.91. We conclude that scale makes the global shift costlier rather than more selective.

Cross-domain transfer. We apply the MMLU-fitted directions, maps, and thresholds to ARC-Challenge and GSM8K-MCQ with no refitting (Table 3), and we plot the two geometries in Fig. 4 (top). ARC-Challenge carries a real overconfident tail and the gated operators transfer: our score-proportional dose reaches +0.342 at $\Delta\text{acc } -0.003$ with retention 0.815, our gated local action +0.290 at $\Delta\text{acc } +0.003$ with retention 0.901, against +0.184 for CAST and +0.151 for MiMiC. The tuned additive dose is higher still at +0.385, and keeps 0.801. GSM8K-MCQ has only 6/300 baseline OCW: the tail targeted on MMLU is absent, our gates read that correctly and fire on nothing, so every conditional operator of ours is the exact identity there ($\Delta\text{acc } 0.000$, retention 1.000), while the same tuned additive dose removes all eleven CR and the tuned CAST gate keeps three of them and costs 6.3 accuracy points. An operator that can abstain pays nothing where the behavior is absent, and an unconditional one cannot tell the difference. Threshold transfer is the part of this that needs care. An absolute MMLU threshold carried to another domain can fire on none of it or on all of it, and which way it errs depends on the axis the threshold sits on. The detector-axis gate at its default quantile over-fires on GSM8K-MCQ and takes CR with it, at retention 0.182 in Table 11; the tuned LDA gate errs the other way and fires on nothing, which is the direction we want an unrefitted threshold to fail in. Label-free re-anchoring of τ to a domain gate-score quantile removes the choice (Appendix J). The sequential gate needs no repair on this domain either, because a gate that reads a prefix and commits once degenerates to the identity where the behavior is absent, which is the behavior the framework asks of it.

Transfer does not push activations off the manifold. A map fitted on one domain and applied to another is where we would expect an activation edit to leave the region the model actually visits. We find it does not,

method (M4, tuned)	MMLU			ARC-Challenge			GSM8K-MCQ		
	Δ_{acc}	ECE	Sel	Δ_{acc}	ECE	Sel	Δ_{acc}	ECE	Sel
baseline	—	0.076	—	—	0.113	—	—	0.258	—
additive CAA (tuned)	+0.000	0.079	+0.232*	+0.003	0.084	+0.385*	−0.030	0.243	+0.000
CAST-style (tuned)	+0.007	0.076	+0.123	+0.007	0.136	+0.184*	−0.063*	0.218	+0.106
MiMiC (tuned)	+0.013	0.110	+0.070	−0.013	0.092	+0.151*	−0.013	0.233	+0.045
Linear-AcT (tuned)	+0.003	0.090	+0.182*	+0.003	0.097	+0.095	+0.023	0.269	+0.227
clamp q_{50}	+0.013	0.101	+0.124	+0.027	0.127	+0.285*	+0.027	0.289	+0.121
gated local action (tuned, ours)	+0.027	0.094	+0.276*	+0.003	0.092	+0.290*	+0.000	0.258	+0.000
gated full-rank transport (tuned, ours)	+0.027	0.090	+0.181*	+0.003	0.105	+0.136	+0.000	0.258	+0.000
score-proportional dose (tuned, ours)	+0.043	0.091	+0.218*	−0.003	0.084	+0.342*	+0.003	0.259	+0.000
sequential gate (tuned, ours)	−0.007	0.114	+0.062	+0.020	0.109	+0.022	+0.000	0.258	+0.000

Table 3. Cross-domain transfer with no refitting: every method keeps the configuration it won on the MMLU tuning split and is applied to ARC-Challenge and GSM8K-MCQ. * marks a bootstrap interval excluding zero. GSM8K-MCQ has 6 baseline OCW and 11 CR, so its selectivity column is noisy by construction, and the accuracy column is not.

provided the edit is projection-local. Conditioning the edited coordinates on the untouched complement gives a statistic that is χ_k^2 under the null, so the fraction of tokens outside the 99% conditional ellipsoid has a value to compare against rather than a threshold to tune. Unsteered traces sit at 0.9–2.9%. The published additive shift puts 100% of tokens outside it on all three domains and the equal-budget dose still puts 40–70% there, while clamping, quantile transport, MiMiC and Linear-AcT stay at 1.0–4.5%, that is at the null (Appendix L). The same ordering appears in the nearest-neighbour distance to a bank of natural layer-14 activations, where the additive shift lands $1.88\times$ further from the model’s own states than the states it replaces on MMLU and further still off-domain, at $2.25\times$ on ARC and $2.04\times$ on GSM8K, and every projection-local operator stays between $1.00\times$ and $1.28\times$. We stress that the edit is small because the target law says it should be, not because we tuned it to be.

Geometry and behavior disagree, and both matter. The two diagnostics do not rank the operators the same way, which is why we report both. Measured by the KL the edit induces on the next-token distribution, the accepted locality criterion in this literature [1], the worst operator is a random direction at the same norm as the additive dose: 0.44 on MMLU against a 0.1 threshold, where the additive dose itself sits at 0.032 and every projection-local operator between 0.000 and 0.006. Yet that random direction is invisible to the conditional test, because it barely projects onto the behavioral coordinates the test watches. The conditional test

model / panel	comparison	Δ calibrated MC2 [adj. 95% CI]	Δ MC1 [adj. 95% CI]	decision
OLMo-2 / TruthfulQA	matched spherical	+ .137 [+ .105, + .168]	+ .105 [+ .025, + .177]	pass
OLMo-2 / TruthfulQA	no intervention	+ .299 [+ .251, + .345]	+ .287 [+ .177, + .384]	pass
OLMo-2 / TruthfulQA	strongest isotropic	+ .220 [+ .180, + .258]	+ .194 [+ .097, + .287]	pass
OLMo-2 / TruthfulQA	unprompted fit	+ .295 [+ .247, + .343]	+ .287 [+ .177, + .384]	pass

Table 4. Held-out OLMo-2 validation on 237 TruthfulQA questions. Intervals use 10,000 paired BCa resamples and the predeclared Bonferroni family. The full seven-control family passes, and the table shows the matched action and representative construction controls. Qwen3 was development-only and is excluded from confirmatory claims.

asks whether the behavioral coordinates were moved to an implausible place and the KL asks whether the next token noticed. An operator has to pass both, and only the projection-local ones do.

Post-review detector-axis validation. We report the positive predeclared held-out check in Table 4. On OLMo-2/TruthfulQA the detector-axis action improves both primary metrics over a norm-matched spherical action and passes all seven pre-frozen construction controls. We can therefore rule out intervention magnitude alone as the explanation, and we find that fitting the representation at prompted answer positions is necessary relative to the unprompted ablation. We read this as evidence that the detector-axis construction is not specific to Gemma.

Qualitative effects. We find gated ablation does not induce generic hedging: hedge markers per trace drop from 0.25 to 0.13 while traces get shorter, from 86 to 71 words. The contrast with the additive dose is not subtle. Under it only 28% of traces still close their <think> block and 20% still emit an answer, distinct-3 falls from 0.91 to 0.31, and 77% of traces are degenerate. Under ours those numbers are unchanged from the unsteered model (Table 5). Samples are in Appendix F, and both lexicons are fixed in advance (Appendix N).

One pass, not two. Theorem 4 lets the same gate decide from a prefix, and the cost of that is measurable rather than assumed. Fitting the LDA head on the first 16 generated positions instead of on the finished trace costs 0.059 of detector AUROC, 0.740 against 0.799 (Appendix G), and the tuning split selects exactly that budget with the threshold at the 30th percentile. The operator then takes one decision per trace, at token 16, and regenerates nothing. Measured end to end on the whole 300-question workload on one idle H100, that is the difference between 45.5 and 85.8

seconds against $43.5\times$ unsteered, or $1.05\times$ against $1.97\times$, because the post-hoc gate must score every finished trace and regenerate the 223 it flags.

Serving cost. Per token and per behavior, we spend one d -dim dot product, a 41-knot lookup, and one FMA ($\approx 4d$ FLOPs $\approx 10^{-4}$ of a layer, the optimal order of Prop. 3); the gate adds a dot product and an $O(1)$ running mean, for a total state of ≈ 19 KB. The action needs no extra forward pass and no KV-cache invalidation. Honesty about what the wall clock does and does not show is worth a sentence. At this scale it does not separate the single-pass operators from each other: every one of them, MiMiC’s $\Theta(d^2)$ map included, lands within 6% of the unsteered time, because the extra arithmetic hides behind a memory-bound decode (Appendix M). What it separates is the number of passes. Strength interpolates as $g_\lambda = (1 - \lambda)\text{id} + \lambda g$, and the operator matches the vLLM residual-stream hook interface [31].

§7. Limitations and Assumptions

The optimality results bound activation displacement under a stated subspace, target law and locality constraint, with the usual regularity conditions on the Monge map. They do not show the projection is sufficient for downstream behavior, and the gate bound uses the concavified ROC, not raw AUROC. Evaluation seeds resample questions under deterministic decoding, not training runs. Architecture transfer is confirmed on OLMo-2/TruthfulQA, while other architecture-behavior cells did not clear every predeclared gate (Appendix A), so we claim no universal detector axis.

§8. Conclusion

We turned an empirical steering failure into theorems and a method. The additive edit is the wrong shape for a tail-compression target, optimal transport along a behavioral subspace is the provably minimal intervention, selectivity is constrained by detection as much as by the action, and the NP-gate $\times$ tail-local action gives the strongest joint trade-off in our Gemma/MMLU comparison across five subsets and two model scales, at no detectable accuracy cost and minimal per-token compute. Two findings we did not expect are worth carrying forward. Equal tuning rehabilitates the additive baseline, and the subspace a full-space operator has over ours turns out not to be where its advantage lives: on a tuning split large enough to identify a configuration, selectivity peaks at $k=2$ and the gated full-rank

member falls to +0.155, so the comparison resolves inside our own framework rather than against it. And the separation between deciding and acting is what makes the method cheap to serve: the decision can be taken on a prefix, with a stated agreement rate against the decision a completed trace would have produced, so nothing is regenerated. We also find that held-out validation on OLMo-2/TruthfulQA supports the detector-axis construction beyond Gemma. We see three directions for further work: better detection axes, transport under a Fisher metric, and further behaviors.

§9. Contributions

Mark Kashirskiy: problem formulation of surgical steering; the PTS framework, Theorems 1–4, Prop. 2 and proofs; the sequential gate and the k -dimensional subspace construction; all steering operators, baseline implementations, and experiments (2B and 9B, five seeds, four benchmarks); visualizations; serving analysis; paper draft. Daniil Sergeev: the confidence measurement suite (M1–M5) and its $n=1000 \times 3$ study; feature-extraction pipeline (CAA/probe/SAE); the original steering grid that exposed the global-hedging failure; the 9B measurement replication; compute infrastructure. Tatevik Ter-Hovhannisyanyan: the proposal of conditional (gated) steering; behavioral labeling methodology and evaluation protocols (jointly with D.S.: the LLM-as-judge and labeled-dataset design for the measurement suite and the next behavioral domains); initial text review of the paper; project direction and coordination. Irina Piontkovskaya (mentor): project direction, research supervision, methodology review; we thank her for guidance throughout the project.

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Appendix §A. Post-review stress-test boundary

The confirmatory program preserved every planned cell in the denominator. Mistral/TruthfulQA improved calibrated MC2 by +.153 with adjusted 95% CI [+.108, +.197], but its MC1 interval [−.059, +.122] crossed

the predeclared joint gate; Granite/TruthfulQA stopped at observer eligibility. Five of six BBQ/ETHICS observer cells did not advance. Mistral/ETHICS did advance and improved pooled calibrated MC2 by $+0.107$ $[+0.090, +0.124]$, but its label-1 MC1 changed by -0.093 $[-0.124, -0.071]$. Thus the OLMo-2 result is a confirmed architecture transfer, while model-wide and behavior-wide generalization remain unsupported. All intervals use 10,000 paired BCa resamples with the predeclared Bonferroni family.

Appendix §B. Full proofs

Theorem 1. By locality, $T(h) - h \in \text{span}(V)$ μ -a.s., hence $\|T(h) - h\|^2 = \|VT(h) - Vh\|^2$ and $C(T) = \mathbb{E}\|VT(h) - Vh\|^2$. Let γ be the joint law of $(Vh, VT(h))$; by the goal constraint γ is a coupling of (μ_V, ν) and $C(T) = \int \|y - x\|^2 d\gamma \geq W_2^2(\mu_V, \nu)$. Since μ_V is absolutely continuous with respect to Lebesgue measure on $\mathbb{R}^k$ with finite second moment (for $k=1$ atomlessness already suffices), Brenier’s theorem gives a unique optimal coupling induced by a map $S = \nabla\varphi$, φ convex; $T^*(h) = h + V^\top(S(Vh) - Vh)$ is admissible and attains the bound. For any optimal T , the induced coupling must equal the unique optimal one, i.e. $VT(h) = S(Vh)$ a.s. Since S depends only on Vh , the optimal intervention is $\sigma(Vh)$ -measurable: access to $h_\perp$ is useless. $\square$

Corollary 1. If $\nu = \mu_V * \delta_\alpha$ the monotone map is $x \mapsto x + \alpha$. Conversely if the optimal map is $x + \alpha$ then ν is the shift of μ_V . For Gaussians with $\sigma_t \neq \sigma_s$, after the best shift the remaining distance is $W_2^2(\mu_V * \delta_\alpha, \nu) = (\sigma_s - \sigma_t)^2 > 0$. Selectivity: let two subpopulations have the same law of p . Any transfer function g then induces the same law of displacements $\Delta = (g(p) - p)\hat{v}$ on both, so no choice of g can address one and spare the other: the operator carries no information that distinguishes them. If in addition the state-change probability depends on h only through (p, Δ) — equivalently $h_\perp$ is conditionally independent of the subpopulation given p — the two marginal change rates coincide and selectivity is zero in expectation. Without that conditional independence a residual gap can survive through $h_\perp$, but it is a property of the downstream geometry and is not reachable by tuning g . Empirically it is $+0.000 \pm 0.000$ at the dose that removes the behavior (Table 11). $\square$

Theorem 2. (a) is the Neyman–Pearson lemma applied to the class-conditional laws of s ; monotone-likelihood-ratio families reduce the LR test to thresholding, and for $\mathcal{N}(\mu_B, \Sigma)$ vs $\mathcal{N}(\mu_G, \Sigma)$ the log-LR is an affine function of the trace feature z , $\log \text{LR}(z) = w^\top z + b$ with $w =$

$\Sigma^{-1}(\mu_B - \mu_G)$; thresholding the scalar $s(z) = w^\top z + b$ is therefore the LR test (LDA). (b) The concavified ROC contains the segment path $(0, 0) \rightarrow (\text{FPR}^*, \text{TPR}^*) \rightarrow (1, 1)$ through the Youden-optimal point; its area is $\frac{1}{2}(1 + \text{TPR}^* - \text{FPR}^*) = \frac{1+J}{2} \leq \text{AUC}$. $\square$

Theorem 3. Decompose $C(T) = P(\Gamma)\mathbb{E}[\|\Delta\|^2 \mid \Gamma] + P(\Gamma^c)\mathbb{E}[\|\Delta\|^2 \mid \Gamma^c]$. Identity off Γ zeroes the second term and is forced by the constraint; the first term with the conditional goal constraint is exactly the Theorem 1 problem for $\mu(\cdot \mid \Gamma)$. The constraints and objective separate over the partition $\{\Gamma, \Gamma^c\}$, so the piecewise optimum is global. $\square$

Proposition 1. The displacement of T_λ is $\lambda(s)^2 \|S(Vh) - Vh\|^2$, so its expectation is $\mathbb{E}_s[\lambda(s)^2 d(s)]$ because λ is s -measurable. The same measurability makes the gain $\mathbb{E}_s[\lambda(s)(\beta_1 \pi(s) - \beta_0(1 - \pi(s)))]$. The objective is therefore an integral over s of $\lambda(\beta_1 \pi - \beta_0(1 - \pi)) - \lambda^2 d/2c$, strictly concave in λ for $d(s) > 0$, and the constraint set $[0, 1]$ is separable in s , so the integral is maximised by maximising the integrand at each s . Setting the derivative to zero gives $\lambda = c((\beta_0 + \beta_1)\pi - \beta_0)/d$ and the clip enforces $[0, 1]$; factoring $\beta_0 + \beta_1$ out of the bracket gives $\pi - \tau$ with $\tau = \beta_0/(\beta_0 + \beta_1)$. Monotonicity in π and the zero set are immediate. As $c/d \rightarrow \infty$ the unclipped value diverges in sign according to $\pi - \tau$, so $\lambda^* \rightarrow \mathbf{1}[\pi > \tau]$ wherever $\pi(s) \neq \tau$, which is P -a.s. when π has no atom at τ ; at any finite c/d the objective at λ^* strictly exceeds its value at the indicator on the event $\{0 < \lambda^* < 1\}$, which then has positive probability. $\square$

Theorem 4. (a) The first t positions generate a σ -algebra $\mathcal{F}_t$; a gate measurable with respect to $\mathcal{F}_t$ is a test between the class-conditional laws of z_t , so Theorem 2(a) applies verbatim and its concavified-ROC bound gives the ceiling $2\text{AUC}(z_t) - 1$. Nothing is assumed about how z_t relates to z_∞ : the head is fitted on z_t itself. (b) Write X_r for the gate contribution of position r and $D_r = X_r - s_\infty$. By assumption (D_r) is a martingale difference sequence with conditionally σ -sub-Gaussian increments, so $\exp(\lambda \sum_{r \leq t} D_r - \lambda^2 \sigma^2 t/2)$ is a nonnegative supermartingale for every λ [10]. Stitching these over geometric epochs and applying Ville's inequality gives the finite-law-of-the-iterated-logarithm boundary and $P(\exists t : |s_t - s_\infty| > B_t(\delta)) \leq \delta$ ([11], Eq. 11 at $\eta=2, s=1.4$). We measure σ rather than assume it: with $\|w\|=1$ and calibration activation norms bounded by R , Hoeffding's lemma makes each contribution R -sub-Gaussian, and we fit the working σ to the deviation profile of the extraction traces. On $\{s_\infty \leq \tau\}$ the firing event $s_t > \tau + B_t(\delta)$ implies $s_t - s_\infty > B_t(\delta)$,

which that bound excludes except on a set of measure δ ; on the complement, $s_t \geq \tau + \Delta - B_t(\delta)$, which exceeds $\tau + B_t(\delta)$ as soon as $B_t(\delta) \leq \Delta/2$, and solving $\sigma\sqrt{2\log(\log 2t/\delta)/t} \leq \Delta/2$ gives $t = O(\sigma^2\Delta^{-2}\log\log(1/\delta))$. (c) The reader is registered on a layer at or above the actor, so within one forward pass the actor for position t observes the state accumulated over positions $< t$; the running mean needs one scalar sum and one counter per sequence, and the edit at position t changes only the keys and values written at position t , which have not yet been consumed. $\square$

Proposition 2. We argue in the latent space rather than through a global change of variables, because a general M does not preserve the duality between the measurement matrix V and the displacement subspace $\text{span}(V^\top)$. Locality writes $\Delta h = V^\top a$ with $a \in \mathbb{R}^k$, and since $VV^\top = I_k$ the goal constraint fixes $a = S(Vh) - Vh$ for whichever map S is used. Hence $\|\Delta h\|_M^2 = a^\top (VMV^\top)a = \|S(q) - q\|_{M_V}^2$ with $q = Vh$: minimising C over admissible T is exactly the Monge problem on $\mathbb{R}^k$ with cost $\|\cdot\|_{M_V}^2$. That cost is a Euclidean square in the coordinates $z = M_V^{1/2}q$ — a change of variables internal to $\mathbb{R}^k$, so the locality constraint is untouched — and Brenier’s theorem in z gives the unique optimal $\nabla\varphi$; mapping back yields $S_M = M_V^{-1/2} \circ \nabla\varphi \circ M_V^{1/2}$. For $M = I$, $M_V = I_k$ and S_M is Theorem 1’s map. $\square$

Proposition 3. Lower bound: let $\hat{v}_j$ be dense ($(\hat{v}_j)_i \neq 0$ for all i) and g non-constant in p_j . If an algorithm leaves coordinate i unread, an adversary perturbs h_i ; this changes $p_j = \langle h, \hat{v}_j \rangle$ and hence, on a set of positive measure, the correct output $T(h)$, while the algorithm’s output is unchanged; contradiction. Learned diff-in-means directions are dense almost surely. Upper bound: PTS computes $p = Vh$ (kd multiply-adds), a monotone K -knot lookup ($O(\log K)$ by binary search), and $h + V^\top(g(p) - p)$ (kd more): $\Theta(kd + \log K)$ total. Memory $\Theta(kd + kK)$: reproducing a target marginal at quantile resolution $1/K$ requires $\Omega(K)$ parameters, and specifying the subspace requires kd . $\square$

Appendix §C. GSM8K MCQ construction

Each GSM8K test item with integer gold g receives three deterministic distractors drawn (seeded by item index) from $\{g \pm 1, g \pm 2, g \pm 10, 2g, \lfloor g/2 \rfloor, \text{round}(1.5g), \text{round}(0.8g), g \pm \text{U}\{3..9\}\}$, deduplicated, shuffled

with the gold into four options. This converts free-form arithmetic into selection while keeping the reasoning burden; we report transfer numbers on it and do not claim GSM8K free-form results.

Appendix §D. The behavior across domains and scales (measurement study)

Overconfidence is model $\times$ instrument (2B). Five confidence readouts on identical MMLU questions ($n=1000 \times 3$ seeds) disagree categorically: ECE ranges from .391 [.374, .409] (letter-logit M2, the literature default) through .249/.177 (self-report/answer-distribution) to .104 [.086, .120] (per-option yes/no M4); disjoint CIs; the instruments even disagree on the answer (all five agree on 28.9% of questions). Hence our protocol requires every steering claim to hold under multiple readouts. Domains: on MMLU the model is overconfident (52/300 OCW), on GSM8K-MC and GPQA underconfident (6 and 27 OCW); the failure mode is domain-dependent, and the input-dependent operators inherit exactly this dependence (Table 3).

Scale replication (9B). The full pipeline was replicated on gemma-2-9b-it (42 layers, $d=3584$, layer 24) over the same domains. Method-dependence survives scale (MMLU ECE .126–.266, GPQA .177–.499, same instrument ranking); scale polarizes rather than cures (GSM8K nearly solved and calibrated, gap -0.015 ; GPQA confidence grows faster than knowledge, M2 conf .86 at acc .36); the confidence cone replicates (cosines 0.63–0.83); confidence becomes more linearly readable (detector AUROC to 0.83 vs 0.62–0.67 at 2B), so the Thm. 2b ceiling rises with scale; and steering remains one-directional, as Remark 1 predicts.

Appendix §E. What each method does: measured movement of real traces

Where the benchmarks live, and how we compare with the label oracle. Figure 4 (left) plots each transfer benchmark’s baseline traces in the same plane: MMLU and ARC carry a real overconfident-wrong mass in the gated (upper-right) region, while GSM8K and GPQA have almost none. That is the geometric reason Table 3 shows activity on the former and near-identity on the latter. Figure 4 (right) compares our intervention to the label oracle, the conditional steering that moves exactly the overconfident-wrong points onto the calibrated centroid: our gate catches 41/50 of the

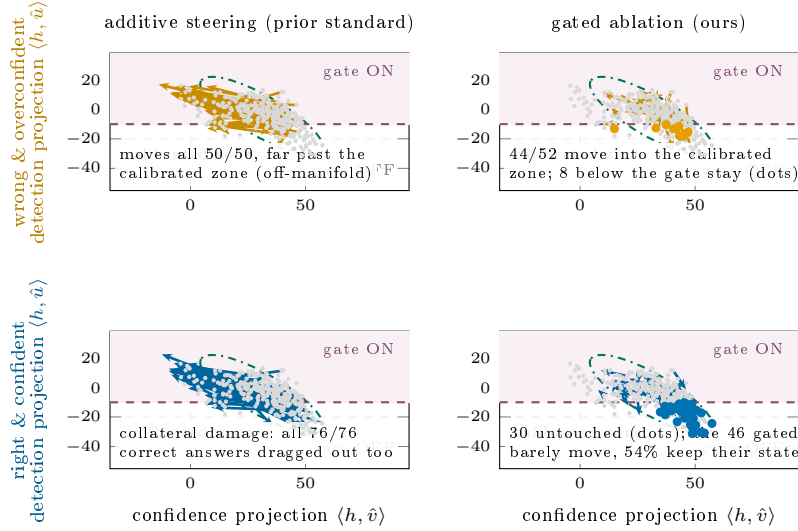


Figure 3. Each dot is one reasoning trace before steering, an arrow is its measured movement after steering on real rollouts, and a bold dot without an arrow was left untouched. The horizontal axis is the confidence projection $\langle h, \hat{v} \rangle$ and the vertical axis the detection projection $\langle h, \hat{u} \rangle$. The pink band is gate ON and the green ellipse is the 2σ contour of the calibrated extraction-split states (Appendix N). Additive steering (left) overshoots all 50 OCW far past the calibrated zone and drags all 76 CR along. Gated ablation (right) moves OCW into the zone and leaves 30 CR bitwise untouched.

oracle’s targets (recall 0.82), misses 9, and false-fires on 46/76 confident-right points (which the tail-local action then mostly spares: 25/46 keep their state). The remaining gap to the oracle is exactly the detection gap of Thm. 2b.

Appendix §F. Sample traces before and after gated ablation

On the 223/300 gated questions, gated ablation does not induce hedging: hedge markers drop from 0.25 to 0.13 per trace and length from 86 to 71 words, 29 of the 50 OCW leave that state and 12 come back correct. Typical outcomes: an associative wrong argument replaced by an explicit

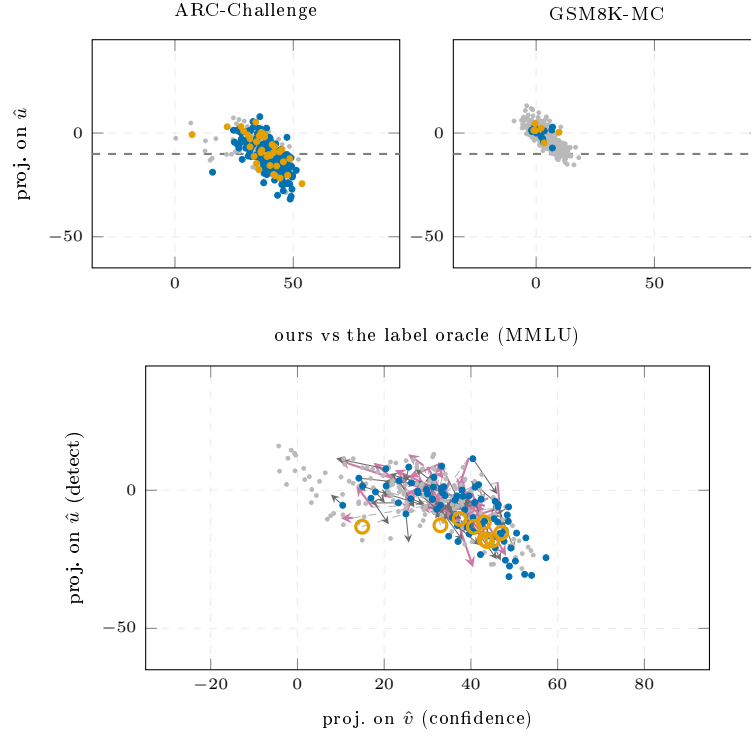


Figure 4. Top: transfer benchmarks in the behavioral plane (gate threshold dashed): ARC has an overconfident-wrong tail above the gate, GSM8K almost none, and transfer activity follows the geometry. Bottom: gray dashed arrows = the label oracle (move every overconfident-wrong point to the calibrated centroid); pink = our measured moves on the same points (45/52 caught); hollow orange circles = the 7 misses; dark-gray thin arrows = false fires on confident-right (mostly state-preserving). The oracle–ours difference is the detection gap that bounds selectivity (Thm. 2b).

derivation, better-calibrated reasoning with the same letter, and a flagged CR rephrased but preserved (strong additive or CAST doses instead yield repetition loops). The excerpts below are truncated, and complete rollouts ship with the released artifact.

condition (MMLU, M2 traces)	words	closes	boxed	distinct-3	degen.%	hedge	certainty
unsteered	86	0.98	0.91	0.91	2.7	0.25	0.12
additive -0.75 [20]	186	0.28	0.20	0.31	76.7	0.08	0.29
clamp q_{50} (ours)	76	0.99	0.94	0.91	2.7	0.18	0.10
gated ablation (ours)	71	0.98	0.95	0.90	3.7	0.13	0.15
sequential gate (ours)	79	0.98	0.93	0.91	2.3	0.24	0.11

Table 5. What each operator does to the reasoning trace. “Closes” is the fraction of traces that emit `</think>` and “boxed” the fraction that emit an answer; degen.% counts traces with distinct-3 below 0.5; hedge and certainty are marker counts per trace from the fixed lexicons of Appendix N.

Across the 223/300 gated questions, the intervention does not inject hedging vocabulary — hedge-marker frequency drops ($0.25 \rightarrow 0.13$ per trace) — and traces get shorter ($86 \rightarrow 71$ words). What changes is the mode of reasoning: assertive shortcut arguments are replaced by definitional, evidence-first ones. Of the 50 overconfident-wrong traces, 29 leave that state and 12 come back correct under M4, while 55 of the 76 confident-right traces keep their state. Ablating the confidence projection during the think phase appears to remove a “commit early” pressure rather than adding doubt. Three representative pairs (traces truncated):
Overconfident-wrong $\rightarrow$ corrected (mmlu-4592; gold C , M4 state overconfident wrong $\rightarrow$ nonconfident wrong). The baseline trace jumps from a vague association to an answer; the steered trace states the definition and derives the (gold) answer.

Question: If the price of firm A’s cell phone service rises by 5 percent and the quantity demanded for firm B’s cell phone service increases by 10 percent, we can say that Options: A. demand for firm B is price elastic. B. supply

before: `<think>`The question asks about the relationship between the price of firm A’s service and the quantity demanded for firm B’s service. We know that the price of firm A’s service increased and the quantity demanded for firm B’s service increased. This suggests that the demand for firm B’s service is likely to be price elastic. `</think>` \boxed{A} $\rightarrow$ A

after: `<think>`The cross-price elasticity of demand measures the percentage change in the quantity demanded of one good in response to a percentage change in the price of another good. Since the price of firm A’s service increased and the quantity demanded for firm B increased, the cross-price elasticity is positive. We can conclude that the two goods are substitutes. `</think>` \boxed{C} $\rightarrow$ C

Overconfident-wrong $\rightarrow$ calibrated-right (mmlu-12923; gold C , M4 state overconfident wrong $\rightarrow$ nonconfident right). The baseline reopens `<think>` three times and re-states the same elimination in each block, so it runs out

of budget before reaching D; the steered trace weighs all four options once and lands on the same letter with calibrated confidence.

Question: The best conclusion that can be drawn about the four-day (compressed) workweek is that it has: Options: A. long-lasting positive effects on attitudes and productivity. B. long-lasting negative effects on attitudes and p

before: <think>The question asks for the best conclusion about the four-day workweek. We need to consider the potential effects on attitudes and productivity. There's no mention of specific evidence or studies, so we need to consider the general impact of a compressed workweek. </think> <think>Option A is too strong. We don't have enough information to say it has long-lasting positive effects on attitudes and productivity. </think> <think>Option B is too strong. We don't have enough information to say it has long-las → D

after: <think>The question asks for the best conclusion about the four-day workweek. We need to consider the potential effects on attitudes and productivity. Option A is too strong, as we don't have enough information to guarantee long-lasting positive effects. Option B is too strong, as we don't have enough information to guarantee long-lasting negative effects. Option C is a possibility, but we don't know if it's the best conclusion. Option D is a possibility, but we don't know if it's the best conclusion. </think> \boxed{D} → D

Confident-right, gated but spared (mmlu-5125; gold B , M4 state confident right → confident right). A false-flagged trace whose state survives the intervention. The state label is the M4 readout, which puts its mass on the gold option B; the displayed rollout is the M2 trace, whose greedy letter is A, so instrument and greedy letter disagree here (Appendix D). Under steering the two agree: the trace states Pavlov's discovery and emits B.

Question: Which one of the following best states Ivan Pavlov's accidental discovery? Options: A. While experimenting with rats, Pavlov discovered that if he repeatedly placed pellets of food in one side of a "T-maze," the rats wou

before: <think>The question asks for Pavlov's accidental discovery. Let's look at the options and see which one describes an accidental discovery. Option B describes a learned response to a stimulus, which is not an accidental discovery. Option C describes a cat learning to press a lever, which is a learned behavior. Option D describes a pigeon learning to associate a key with food. Option A describes a rat learning to run to a specific side of a maze. This is a classic example of classical conditioning. Therefore, → A

after: <think>The question asks for the accidental discovery of Pavlov. Option B is the best choice because it describes Pavlov's accidental discovery of the dog's salivation in response to the sight and sound of the attendant who brought them food. </think> \boxed{B} → B

One question, four methods. The most instructive comparison holds the question fixed (mmlu-4592, gold C; baseline answers A, overconfident-wrong). Our gated ablation produces the clean definitional trace shown

above and lands on C. The additive-action methods do something qualitatively different — their traces degenerate into verbatim repetition loops, the signature of support violation (Remark 6): the constant shift pushes activations off the data manifold and generation cycles. Trace length balloons from 351 to ~ 1500 characters of looped text; this is not deliberation but decoding pathology, invisible in aggregate confidence numbers yet obvious in the rollouts. MiMiC (distribution-matched, unconditional) stays fluent but keeps the wrong answer.

additive -0.75 [20] ($\rightarrow$ A):

<think>The cross-price elasticity of the cell phone service is equal to the price of the cell phone service of the firm A and the quantity of the cell phone service of the firm B. The cross-price elasticity of the cell phone service is equal to the price of the cell phone service of the firm A and the quantity of the cell phone service of

CAST-style gate+add [13] ($\rightarrow$ A):

<think>The cross-price elasticity of demand for firm B is $(10/5) = 2$. The cross-price elasticity of demand for firm A is $(10/5) = 2$. The cross-price elasticity of demand for firm A is $(10/5) = 2$. The cross-price elasticity of demand for firm B is $(10/5) = 2$. The cross-price elasticity of demand for firm A is $(10/5) = 2$. The cross-price el

MiMiC [24] ($\rightarrow$ A):

<think>The question asks about the relationship between the price of firm A's service and the quantity demanded for firm B's service. We know that the price of firm A's service increased and the quantity demanded for firm B's service increased. This suggests that the demand for firm B's service is

gated ablation (ours) ($\rightarrow$ C):

<think>The cross-price elasticity of demand measures the percentage change in the quantity demanded of one good in response to a percentage change in the price of another good. Since the price of firm A's service increased and the quantity demanded for firm B increased, the cross-price elasticity is positive. We can conclude that the tw

Appendix §G. Where to read and where to act: the layer sweep

Table 6 refits both axes and the gate at nine depths of gemma-2-2b-it on the same 400 extraction traces. Two quantities move in opposite directions. Token-level confidence separation grows monotonically with depth (Cohen's d from 0.53 to 0.70), because late layers carry the answer itself. Detector quality, the quantity Theorem 2b turns into a selectivity ceiling, peaks in the middle and collapses in the last layers, where the residual stream has already committed. Layer 14 is inside the plateau and layer 16 is its maximum; the gap between them, $2\text{AUROC} - 1$ of 0.538 against

layer	Cohen's d conf.	Cohen's d OCW/CR	gate AUROC	ceiling 2AUC-1
4	+0.525	-0.079	0.783	+0.566
7	+0.535	-0.093	0.756	+0.511
10	+0.512	-0.144	0.741	+0.482
12	+0.516	-0.150	0.742	+0.485
14	+0.589	-0.155	0.769	+0.538
16 $\star$	+0.631	-0.141	0.799	+0.599
18	+0.616	-0.116	0.779	+0.558
22	+0.675	-0.110	0.791	+0.582
25	+0.700	-0.111	0.649	+0.298

Table 6. Layer sweep on the extraction split (400 traces, both axes and the LDA gate refit at each depth). $\star$ marks the depth the sequential gate reads. The OCW/CR column is measured on the axis refit at that depth and is therefore not the $d = 0.049$ of Fig. 1, which is measured on the layer-14 axis the operators act through.

0.599, is why the sequential gate reads two layers above the layer it edits (Remark 1).

What a prefix budget costs the gate. A head fitted on completed traces is a detector, and a detector is not automatically a predictor: features that recognise a behavior in finished text are known to forecast it poorly from a prefix [5]. We therefore do not reuse the whole-trace head online. For each budget t we refit the LDA on the mean projection over the first t generated positions and measure its AUROC on the same contrast (Table 7). The cost of deciding early is smaller than the worry suggests: at layer 16 a head that sees eight generated tokens reaches 0.736 against 0.799 for the whole trace, and by 128 tokens the gap is 0.016. The dip at $t=32$ is sampling noise in a two-feature fit on 400 traces and does not survive at either neighbour. Layer 16 dominates layer 14 at every budget, which is why the sequential gate reads there.

Appendix §H. Equal-budget tuning

Every method in Table 1 receives at least as many configurations as our own eight, on the same held-out MMLU tuning split of $n=1200$ questions, disjoint from the extraction split and from all five evaluation subsets, and we select among them by one pre-declared rule: the highest selectivity

gate AUROC (OCW vs CR)	$t=8$	16	32	64	128	256
layer 14	0.708	0.712	0.666	0.707	0.747	0.766
layer 16	0.736	0.740	0.691	0.738	0.783	0.795

Table 7. Detector quality as a function of the generated-token budget the gate is allowed to see, with the head refitted for each budget. The whole-trace values are 0.769 at layer 14 and 0.799 at layer 16.

among configurations whose accuracy change is at least -0.01 . The additive baseline searches its dose, CAST searches threshold $\times$ dose, MiMiC searches layer $\times$ shrinkage, Linear-AcT searches layer $\times \lambda$, our conditional ablation searches the depth the gate reads $\times$ the quantile it fires at $\times$ the local action, our transport variant searches the subspace dimension $k \times$ the gate quantile, and the sequential gate searches its prefix budget $t^* \times$ the gate quantile. We report each winner in Table 8, and those configurations, unchanged, produce the evaluation-seed numbers in the main text. The split has to be that large. On a 400-question split six of the nine winners are different and the measured selectivity of a fixed configuration moves by up to 0.131 (Table 9), so the smaller split does not identify a configuration, it samples one. Every search also runs at one batch size, because batched greedy decoding is not invariant to it and the same MiMiC configuration measures $+0.288$ at 64 and $+0.235$ at 32. Each method’s search is paired against a baseline generated in the same process, which is what config selection needs; the searches ran on different GPUs and their baselines differ by up to three OCW out of 79, so we do not compare selectivities across methods on this split. That comparison is Table 1, where every method is paired against one shared set of baseline rollouts per seed. We also report one configuration that is not part of any selection, the NP gate composed with the full-space affine map, because Theorem 3 predicts the gate composes with any transport and that is the cheapest way to test the prediction on the strongest unconditional action.

Appendix §I. Design ablations of the operator family

Every member of the family Theorem 1 defines was tuned under the same budget and carried unchanged to the five evaluation subsets. The point of the table is which design choices survive that carry. Raising the subspace dimension does not: the tuned k falls to $+0.065$ on the subsets

method	budget	selected configuration	Δ_{acc}	ocw_rm	cr_keep	Sel
additive CAA [20]	8	alpha-0.25	+0.016	0.627	0.644	+0.271
CAST-style [13]	8	tau_cr_q70_alpha-0.375	-0.005	0.428	0.756	+0.184
MiMiC [24]	9	L18_reg0.1	+0.030	0.468	0.795	+0.263
Linear-AcT [21]	9	L14_lam1.0	+0.026	0.318	0.865	+0.184
gated local action (ours)	8	gl14_q30_ablate	+0.001	0.527	0.718	+0.245
gated k -dim transport (ours)	8	k2_q30	+0.005	0.279	0.907	+0.186
score-proportional dose (ours)	8	gl16_q10_kappa1.0	+0.010	0.607	0.660	+0.267
sequential gate (ours, single pass)	8	q50_at32	+0.014	0.219	0.917	+0.136
NP-gate $\times$ MiMiC (Thm. 3)	4	L14_q30_mimic	+0.022	0.318	0.837	+0.155

Table 8. Equal-budget hyperparameter search on the tuning split.

method	$n=400$ split		$n=1200$ split		
	winner	Sel	winner	Sel	Δ
additive CAA [20]	alpha-0.25	+0.196	alpha-0.25	+0.271	+0.075
CAST-style [13]	tau_cr_q70_alpha-0.75	+0.182	tau_cr_q70_alpha-0.375	+0.184	+0.003
MiMiC [24]	L18_reg0.01	+0.288	L18_reg0.1	+0.263	-0.025
Linear-AcT [21]	L10_lam0.25	+0.099	L14_lam1.0	+0.184	+0.085
gated local action (ours)	gl14_q30_ablate	+0.232	gl14_q30_ablate	+0.245	+0.013
gated k -dim transport (ours)	k32_q30	+0.165	k2_q30	+0.186	+0.021
score-proportional dose (ours)	gl16_q10_kappa0.5	+0.265	gl16_q10_kappa1.0	+0.267	+0.003
sequential gate (ours, single pass)	q30_at16	+0.120	q50_at32	+0.136	+0.015
NP-gate $\times$ MiMiC (Thm. 3)	L18_q30_mimic	+0.286	L14_q30_mimic	+0.155	-0.131

Table 9. The same search on a tuning split three times larger. Six of the nine winners change and the measured selectivity moves by up to 0.131, so a 400-question split does not identify a configuration reliably. The paper uses the $n=1200$ winners throughout.

from +0.186 on the split. Making the action token-local rather than trace-local does not either, +0.158 from +0.257, and neither does weighting the dose by a posterior. Grading the dose by the gate margin does survive, +0.231 from +0.267, and so does the plain $k=2$ gated ablation, +0.206 from +0.245, which loses the least of anything we tried. One member is worth reporting on its own terms. Theorem 3 gives the cost-minimal gated intervention as the transport map of the law conditioned on the gate event, and every gated transport above uses the marginal law instead. We fitted the conditional moments on the extraction split, at each of the four gate quantiles, and compared the two maps under one budget. The conditional map is not better: +0.155 against +0.183 at the 20th percentile, +0.153 against +0.125 at the 30th, and no systematic ordering across the four. We read that as Theorem 3 delivering what it claims and nothing else. It minimises displacement subject to reaching the target law, which is not

member (M4, MMLU, seeds mean $\pm$ std)	Δ acc	ECE	cr_keep	Sel
baseline	acc 0.473 $\pm$ 0.026	0.099 $\pm$ 0.019	—	—
gated ablation, $k=2$	-0.015 $\pm$ 0.036	0.089 $\pm$ 0.013	0.713 $\pm$ 0.050	+0.206 $\pm$ 0.059
gated transport, tuned k	-0.010 $\pm$ 0.012	0.100 $\pm$ 0.011	0.882 $\pm$ 0.053	+0.065 $\pm$ 0.095
gated token-local clamp	-0.008 $\pm$ 0.053	0.095 $\pm$ 0.023	0.721 $\pm$ 0.050	+0.158 $\pm$ 0.085
score-proportional dose	-0.005 $\pm$ 0.054	0.089 $\pm$ 0.010	0.650 $\pm$ 0.056	+0.231 $\pm$ 0.120
posterior-weighted dose	-0.003 $\pm$ 0.006	0.093 $\pm$ 0.017	0.969 $\pm$ 0.027	+0.042 $\pm$ 0.050
gated full-rank transport ($k=d$)	+0.003 $\pm$ 0.021	0.108 $\pm$ 0.022	0.854 $\pm$ 0.038	+0.146 $\pm$ 0.042

Table 10. Members of the projection-transport family under the equal-budget protocol, each at the configuration it won on the $n=1200$ tuning split.

method (M4, MMLU, seeds mean $\pm$ std)	Δ acc	ECE	AURC	Sel
baseline	acc 0.473 $\pm$ 0.026	0.099 $\pm$ 0.019	0.404 $\pm$ 0.029	—
prompting	-0.005 $\pm$ 0.020	0.118 $\pm$ 0.018	0.392 $\pm$ 0.023	+0.198 $\pm$ 0.086
additive -0.75 [20]	-0.087 $\pm$ 0.053	0.108 $\pm$ 0.044	0.556 $\pm$ 0.056	+0.000 $\pm$ 0.000
CAST-style gate+add [13]	-0.049 $\pm$ 0.039	0.099 $\pm$ 0.021	0.450 $\pm$ 0.040	+0.222 $\pm$ 0.055
MiMiC full-space affine [24]	-0.001 $\pm$ 0.041	0.116 $\pm$ 0.026	0.401 $\pm$ 0.048	+0.200 $\pm$ 0.107
Linear-AcT $\lambda=0.5$ [21]	-0.006 $\pm$ 0.036	0.104 $\pm$ 0.015	0.412 $\pm$ 0.018	+0.120 $\pm$ 0.080
Linear-AcT $\lambda=1.0$ [21]	+0.001 $\pm$ 0.032	0.101 $\pm$ 0.011	0.410 $\pm$ 0.027	+0.146 $\pm$ 0.141
clamp q_{50} (ours)	+0.001 $\pm$ 0.034	0.110 $\pm$ 0.033	0.408 $\pm$ 0.020	+0.145 $\pm$ 0.039
quantile-OT non-OCW (ours)	-0.001 $\pm$ 0.029	0.108 $\pm$ 0.017	0.408 $\pm$ 0.015	+0.119 $\pm$ 0.086
gated clamp (ours)	-0.001 $\pm$ 0.039	0.096 $\pm$ 0.021	0.404 $\pm$ 0.024	+0.224 $\pm$ 0.065
gated ablation (ours)	-0.003 $\pm$ 0.050	0.094 $\pm$ 0.030	0.395 $\pm$ 0.038	+0.279 $\pm$ 0.090
LDA-gated ablation (ours)	-0.009 $\pm$ 0.032	0.091 $\pm$ 0.010	0.402 $\pm$ 0.028	+0.198 $\pm$ 0.094
online-gated ablation (ours, single pass)	-0.000 $\pm$ 0.016	0.109 $\pm$ 0.016	0.405 $\pm$ 0.040	+0.035 $\pm$ 0.048

Table 11. The same five evaluation subsets with every method at its default configuration, that is before the equal-budget search of Appendix H.

the same objective as maximising selectivity, and selectivity here is bound by the detector rather than by the map (Thm. 2b). Five attempts to buy more removal by changing the action — a larger subspace, a full-rank map, a graded dose, a token-local clamp and the conditional map — each bought it at the same rate in retention, which is what Cor. 1 predicts for a subpopulation pair separated by $d = 0.049$.

We report the members that did not carry because each one is an insight about the selection problem rather than a catalogue entry. Together they say that the more exposure or dose a configuration buys on a tuning split, the less of it survives a change of questions, and that the design choice which survives best is the one the theory singles out: a two-dimensional edit applied behind a gate.

condition (M4 readout)	Δacc	95% CI	ECE	OCW left	cr_keep	Sel [95% CI]
baseline	—	—	0.076	50	—	—
additive -0.75						
citerimsky2024caa	-0.150	[-0.21, -0.08]	0.052	0	0.000	+0.000 [+0.00, +0.00]
ablation						
citearditi2024refusal	+0.020	[-0.04, +0.08]	0.079	30	0.592	+0.232 [+0.07, +0.40]
clamp q_{50}	+0.013	[-0.04, +0.07]	0.101	48	0.724	+0.124 [-0.05, +0.30]
quantile-OT non-OCW	-0.007	[-0.04, +0.03]	0.096	54	0.829	+0.029 [-0.12, +0.18]
gated clamp q_{50}	+0.017	[-0.03, +0.07]	0.081	47	0.855	+0.195 [+0.04, +0.35]
gated clamp q_{70}	-0.003	[-0.04, +0.04]	0.045	39	0.895	+0.175 [+0.04, +0.32]
gated ablation	+0.033	[-0.02, +0.09]	0.101	35	0.724	+0.304 [+0.13, +0.47]
sequential-gate ablation	+0.003	[-0.03, +0.04]	0.098	49	0.934	+0.074 [-0.03, +0.19]

Table 12. MMLU, $n=300$, layer 14, M4 readout (the calibrated instrument). The additive baseline removes all OCW by destroying CR (selectivity ≈ 0 , as Cor. 1 forces). Input-dependent transfer functions achieve positive selectivity at an accuracy gain. Same-sign effects hold under the M2 readout (Appendix J).

Appendix §J. Full result tables

Full condition grids under both readouts. Columns: condition, readout, Δ accuracy with paired-bootstrap 95% CI, ECE, count of OCW after steering, CR-retention, selectivity with 95% CI. subbw = ungated subspace Bures–Wasserstein transport; npbw = LDA-gated subspace transport (self_qXX = gate threshold re-anchored, label-free, to the target domain’s own score quantile); -think = steering active only during the <think> phase.

Appendix §K. Capability outside the target behavior

The operators are fitted on multiple-choice reasoning traces, so the question is what they do to generation that has nothing to do with the target behavior. Table 16 reports code synthesis (HumanEval pass@1, greedy, $n=164$), held-out language modelling (WikiText-103 perplexity, $n=300$ passages), and open-ended instruction following (200 AlpacaEval prompts, 384 new tokens): fluency as the perplexity the unsteered model assigns to the continuation, repetition as distinct-3, and length in words. The last column is the fraction of prompts on which the sequential gate fires at all.

Appendix §L. Do steered activations stay on the manifold?

We teacher-force the unsteered traces, apply each operator to the layer-14 residuals, and measure how far the result travels. Table 17 reports the relative displacement $\|h' - h\|/\|h\|$, the 99th percentile of the norm ratio $\|h'\|/\|h\|$, and the distance to the nearest natural activation in a bank of 4,096 unsteered residuals relative to the unsteered nearest-neighbour distance. We include the transfer domains ARC and GSM8K because that is where a map fitted on MMLU could be expected to explode.

One negative result is worth stating rather than hiding. A full-space Mahalanobis distance is useless here: an operator that edits a k -dimensional projection leaves $d - k$ coordinates untouched, which dilute the statistic until it moves by under 1% for every operator in the table, including the one that destroys generation. The same dilution has been reported independently [30]. We therefore condition instead. Writing $h = (q, h_\perp)$ with $q = Vh$, the law of q given the 64 leading principal directions of the complement is Gaussian under the layer’s fitted moments, and its squared Mahalanobis distance is χ_k^2 under the null, so the natural readout is the fraction of tokens outside the 99% conditional ellipsoid, against an expected 1%. We also report the behavioral check the refusal literature uses as a selection criterion, the mean KL between the unsteered and steered next-token distributions with 0.1 as the accepted threshold [1], measured against a norm-matched random direction so that the number is interpretable [29]. We claim only that a projection-local edit lands closer to the states the model produces by itself, not that it lands on them: steered states are provably unreachable from any discrete prompt [17].

Appendix §M. End-to-end serving cost

Table 18 measures the whole 300-question workload on one H100, so the post-hoc gate is charged for its scoring pass and for regenerating what it flags.

Appendix §N. Definitions used in the text

Hedge markers. The lexicon counted in Sec. 5 and Appendix F is fixed in advance and case-insensitive: maybe, perhaps, might, possibly, however, wait, actually, hmm, unsure, unclear, guess, seems, probably, could, uncertain, confused. The matching certainty lexicon is clearly, obviously,

definitely, certainly, surely, undoubtedly, evidently, must, certain, sure. A trace’s hedge count is the number of token matches; rates are per trace.

Calibrated zone. The green ellipse in Fig. 3 is the 2σ contour of the empirical Gaussian fitted to the trace-mean projections of the calibrated extraction-split states (CR and nonconfident-wrong), i.e. the states the transport target is built from. It is a visual reference, not a decision boundary.

Norm-matched spherical action. The comparison action in Table 4 is a spherical steering operator: activations are rotated along the fixed-azimuth shortest geodesic towards a unit prototype at constant norm, with the decision taken by an antipodal von Mises–Fisher confidence at concentration $\kappa=20$. Its nine-cell grid ($\alpha \in \{0.6, 0.9, 1\}$, $\beta \in \{0.6, 0.9, 0.99\}$) is selected on the observer split, and the reported comparison uses the selected cell, so the two conditions differ in the direction and the decision rule and not in how far they move activations. The five isotropic controls replace the detector direction with a random unit direction at exactly the measured update norm; the unprompted control refits the direction at last-token positions without the evaluation instruction. All are input-independent given the fit.

Condition names. The exhaustive grids in Appendix J carry the run identifiers; the main text uses short names. `m4_conf_*` is an action on the confidence axis; `sweep_ocwcr-crq50_ablate` is gated ablation (gate fitted on the OCW-versus-CR contrast, threshold at the CR median); `sweep_lda-allq50_ablate` is LDA-gated ablation (the two-dimensional LDA head of Sec. 3, threshold at the median of all extraction scores); `online_d0p05_q50_L16_ablate` is its sequential single-pass form at $\delta=0.05$; `probegate-*` gates on a full- d logistic probe; `cast_gate-cr_q50_add-0.75` is the CAST-style reimplementation; `act_l05/act_l10` are Linear-AcT at $\lambda=0.5/1.0$; `mimic_full` is MiMiC; `subspace-BW` is ungated subspace Bures–Wasserstein transport and `NP-gate×BW` its gated form (`subbw` and `npbw` in the run identifiers); `self_qXX` marks a threshold re-anchored, label-free, to the target domain’s own score quantile; and `-think` marks steering confined to the `<think>` phase.

condition	r/o	$\Delta\text{acc [CI]}$	ECE	OCW	cr_keep	Sel [CI]
baseline	m4	—	0.076	50	—	—
Linear-AcT $\lambda=0.5$	m4	−0.007 [−0.053, +0.037]	0.084	50	0.83	+0.109 [−0.04, +0.25]
Linear-AcT $\lambda=1.0$	m4	+0.003 [−0.037, +0.043]	0.090	48	0.88	+0.182 [+0.03, +0.33]
CAST-style gate+add	m4	−0.067 [−0.130, −0.003]	0.079	9	0.39	+0.215 [+0.05, +0.37]
ablation	m4	+0.020 [−0.040, +0.080]	0.079	30	0.59	+0.232 [+0.07, +0.40]
additive −0.75	m4	−0.150 [−0.213, −0.080]	0.052	0	0.00	+0.000 [+0.00, +0.00]
clamp q_{30}	m4	+0.027 [−0.030, +0.083]	0.102	40	0.68	+0.184 [+0.01, +0.36]
clamp q_{50}	m4	+0.013 [−0.043, +0.070]	0.101	48	0.72	+0.124 [−0.05, +0.30]
clamp q_{70}	m4	+0.023 [−0.030, +0.077]	0.136	48	0.72	+0.144 [−0.03, +0.31]
gated clamp q_{50}	m4	+0.017 [−0.033, +0.067]	0.081	47	0.86	+0.195 [+0.04, +0.35]
gated OT q_{50}	m4	−0.023 [−0.067, +0.020]	0.095	50	0.84	+0.042 [−0.10, +0.18]
gated clamp q_{70}	m4	−0.003 [−0.043, +0.040]	0.045	39	0.89	+0.175 [+0.04, +0.32]
gated OT q_{70}	m4	−0.010 [−0.037, +0.017]	0.083	53	0.92	+0.001 [−0.09, +0.10]
Gaussian-OT cal.	m4	+0.027 [−0.010, +0.067]	0.119	48	0.88	+0.062 [−0.06, +0.19]
quantile-OT cal.	m4	+0.007 [−0.030, +0.047]	0.131	55	0.84	+0.002 [−0.13, +0.14]
quantile-OT non-OCW	m4	−0.007 [−0.043, +0.030]	0.096	54	0.83	+0.029 [−0.12, +0.18]
MiMiC full-space affine	m4	+0.013 [−0.037, +0.063]	0.098	43	0.83	+0.249 [+0.09, +0.40]
sequential-gate ablation	m4	+0.003 [−0.027, +0.037]	0.098	49	0.93	+0.074 [−0.03, +0.19]
online_d0p05_q50_L16_t16_ablate	m4	+0.017 [−0.023, +0.057]	0.124	53	0.79	−0.011 [−0.15, +0.13]
online_d0p05_q50_L16_t64_ablate	m4	−0.007 [−0.027, +0.013]	0.075	47	0.93	+0.054 [−0.05, +0.16]
proagate-all_q60_ablate	m4	+0.013 [−0.023, +0.053]	0.105	42	0.74	+0.117 [−0.04, +0.29]
proagate-all_q60_clamp_q30	m4	+0.027 [−0.010, +0.063]	0.109	46	0.78	+0.076 [−0.08, +0.24]
proagate-all_q60_otq_cal	m4	−0.013 [−0.043, +0.017]	0.088	48	0.92	+0.061 [−0.05, +0.19]
proagate-all_q75_ablate	m4	+0.020 [−0.013, +0.057]	0.107	44	0.87	+0.188 [+0.04, +0.34]
proagate-all_q75_clamp_q30	m4	+0.033 [+0.000, +0.070]	0.074	40	0.86	+0.195 [+0.05, +0.35]
proagate-all_q75_otq_cal	m4	−0.003 [−0.030, +0.027]	0.089	51	0.89	+0.035 [−0.08, +0.16]
prompting	m4	−0.017 [−0.073, +0.043]	0.096	58	0.70	+0.097 [−0.08, +0.26]
LDA-gated ablation	m4	+0.020 [−0.013, +0.053]	0.079	38	0.83	+0.209 [+0.04, +0.37]
sweep_lda-allq50_clamp_q30	m4	+0.030 [−0.003, +0.067]	0.093	44	0.89	+0.175 [+0.03, +0.33]
sweep_lda-allq50_clamp_q50	m4	+0.013 [−0.020, +0.047]	0.100	45	0.95	+0.187 [+0.06, +0.33]
gated ablation	m4	+0.033 [−0.020, +0.090]	0.101	35	0.72	+0.304 [+0.13, +0.47]
sweep_ocw-cr-crq50_clamp_q30	m4	+0.007 [−0.040, +0.053]	0.078	48	0.84	+0.222 [+0.07, +0.38]
gated clamp q_{50} (sweep)	m4	+0.017 [−0.033, +0.067]	0.081	47	0.86	+0.195 [+0.04, +0.35]
Linear-AcT (tuned)	m4	+0.003 [−0.037, +0.043]	0.090	48	0.88	+0.182 [+0.03, +0.33]
additive CAA (tuned)	m4	+0.000 [−0.057, +0.063]	0.079	40	0.59	+0.232 [+0.05, +0.41]
CAST-style (tuned)	m4	+0.007 [−0.037, +0.053]	0.076	33	0.76	+0.123 [−0.04, +0.29]
gated full-rank transport (tuned, ours)	m4	+0.027 [−0.010, +0.063]	0.090	46	0.92	+0.181 [+0.05, +0.32]
MiMiC (tuned)	m4	+0.013 [−0.040, +0.070]	0.110	56	0.75	+0.070 [−0.09, +0.23]
sequential gate (tuned, ours)	m4	−0.007 [−0.037, +0.023]	0.114	49	0.88	+0.062 [−0.06, +0.19]
gated local action (tuned, ours)	m4	+0.027 [−0.020, +0.077]	0.094	35	0.78	+0.276 [+0.11, +0.45]
gated k -dim transport (tuned, ours)	m4	−0.027 [−0.063, +0.013]	0.095	57	0.82	−0.064 [−0.19, +0.06]
tuned_ourspost	m4	−0.003 [−0.020, +0.013]	0.075	48	0.99	+0.067 [−0.01, +0.15]
score-proportional dose (tuned, ours)	m4	+0.043 [−0.010, +0.100]	0.091	33	0.66	+0.218 [+0.04, +0.39]
gated token-local clamp (tuned, ours)	m4	+0.023 [−0.023, +0.070]	0.084	39	0.76	+0.263 [+0.10, +0.44]
baseline	m2	—	0.375	124	—	—
Linear-AcT $\lambda=0.5$	m2	+0.017 [−0.013, +0.047]	0.360	121	0.94	+0.054 [−0.01, +0.12]
Linear-AcT $\lambda=1.0$	m2	+0.007 [−0.027, +0.040]	0.374	124	0.92	+0.052 [−0.02, +0.12]
CAST-style gate+add	m2	−0.110 [−0.167, −0.050]	0.227	89	0.60	+0.090 [−0.03, +0.20]
ablation	m2	+0.013 [−0.037, +0.063]	0.344	120	0.84	+0.106 [+0.01, +0.20]
additive −0.75	m2	−0.193 [−0.260, −0.127]	0.219	98	0.38	−0.051 [−0.16, +0.07]
clamp q_{30}	m2	+0.010 [−0.040, +0.057]	0.354	122	0.85	+0.104 [+0.01, +0.20]
clamp q_{50}	m2	+0.007 [−0.037, +0.050]	0.368	125	0.88	+0.059 [−0.02, +0.14]
clamp q_{70}	m2	+0.003 [−0.043, +0.047]	0.380	126	0.87	+0.063 [−0.02, +0.15]
gated clamp q_{50}	m2	+0.023 [−0.020, +0.067]	0.355	119	0.90	+0.101 [+0.02, +0.18]
gated OT q_{50}	m2	+0.000 [−0.027, +0.027]	0.380	127	0.95	+0.009 [−0.04, +0.06]
gated clamp q_{70}	m2	+0.030 [−0.007, +0.067]	0.348	116	0.95	+0.116 [+0.04, +0.19]
gated OT q_{70}	m2	+0.010 [−0.007, +0.027]	0.370	121	0.99	+0.045 [+0.00, +0.09]
Gaussian-OT cal.	m2	−0.003 [−0.030, +0.023]	0.380	128	0.95	−0.005 [−0.05, +0.04]
quantile-OT cal.	m2	+0.010 [−0.017, +0.040]	0.368	125	0.95	+0.035 [−0.02, +0.10]
quantile-OT non-OCW	m2	+0.017 [−0.010, +0.047]	0.370	122	0.96	+0.055 [−0.00, +0.12]
MiMiC full-space affine	m2	+0.020 [−0.023, +0.063]	0.360	120	0.88	+0.091 [+0.01, +0.17]
sequential-gate ablation	m2	+0.007 [−0.010, +0.027]	0.363	123	0.98	+0.017 [−0.02, +0.06]
sequential-gate clamp q_{50}	m2	+0.000 [−0.017, +0.017]	0.377	124	0.98	+0.006 [−0.03, +0.04]
online_d0p05_q50_L16_t16_ablate	m2	+0.013 [−0.010, +0.040]	0.357	120	0.96	+0.053 [−0.00, +0.11]
online_d0p05_q50_L16_t64_ablate	m2	+0.010 [−0.007, +0.027]	0.369	121	0.99	+0.029 [−0.01, +0.07]
proagate-all_q60_ablate	m2	+0.000 [−0.033, +0.030]	0.367	122	0.93	+0.048 [−0.02, +0.12]
proagate-all_q60_clamp_q30	m2	−0.013 [−0.047, +0.020]	0.383	127	0.92	+0.020 [−0.04, +0.09]
proagate-all_q60_otq_cal	m2	+0.013 [−0.007, +0.033]	0.369	120	0.98	+0.047 [+0.00, +0.10]
proagate-all_q75_ablate	m2	+0.000 [−0.030, +0.030]	0.365	123	0.93	+0.018 [−0.04, +0.08]
proagate-all_q75_clamp_q30	m2	−0.007 [−0.037, +0.023]	0.378	126	0.92	+0.012 [−0.05, +0.08]
proagate-all_q75_otq_cal	m2	+0.007 [−0.017, +0.030]	0.367	123	0.97	+0.027 [−0.02, +0.07]
prompting	m2	+0.020 [−0.030, +0.073]	0.344	119	0.85	+0.110 [+0.01, +0.20]
LDA-gated ablation	m2	−0.007 [−0.033, +0.020]	0.365	125	0.94	+0.005 [−0.05, +0.06]
sweep_lda-allq50_clamp_q30	m2	+0.003 [−0.023, +0.030]	0.366	120	0.95	+0.043 [−0.02, +0.11]
sweep_lda-allq50_clamp_q50	m2	+0.003 [−0.020, +0.030]	0.365	123	0.96	+0.023 [−0.03, +0.08]
gated ablation	m2	+0.030 [−0.013, +0.073]	0.337	115	0.89	+0.119 [+0.03, +0.21]
sweep_ocw-cr-crq50_clamp_q30	m2	+0.020 [−0.020, +0.063]	0.347	117	0.91	+0.129 [+0.05, +0.21]
gated clamp q_{50} (sweep)	m2	+0.023 [−0.020, +0.067]	0.355	119	0.90	+0.101 [+0.02, +0.18]
Linear-AcT (tuned)	m2	+0.007 [−0.027, +0.040]	0.374	124	0.92	+0.052 [−0.02, +0.12]
additive CAA (tuned)	m2	+0.007 [−0.043, +0.053]	0.330	117	0.83	+0.108 [+0.02, +0.20]
CAST-style (tuned)	m2	+0.007 [−0.033, +0.047]	0.296	103	0.87	+0.152 [+0.06, +0.25]
gated full-rank transport (tuned, ours)	m2	+0.000 [−0.030, +0.030]	0.369	123	0.92	+0.036 [−0.03, +0.11]
MiMiC (tuned)	m2	+0.030 [−0.013, +0.073]	0.352	117	0.91	+0.123 [+0.04, +0.21]
sequential gate (tuned, ours)	m2	+0.007 [−0.020, +0.033]	0.365	122	0.95	+0.033 [−0.02, +0.09]
gated local action (tuned, ours)	m2	+0.003 [−0.033, +0.043]	0.357	121	0.90	+0.053 [−0.02, +0.13]
gated k -dim transport (tuned, ours)	m2	−0.007 [−0.033, +0.017]	0.379	125	0.95	+0.011 [−0.04, +0.07]
tuned_ourspost	m2	−0.003 [−0.017, +0.007]	0.374	125	0.99	−0.004 [−0.03, +0.02]
score-proportional dose (tuned, ours)	m2	−0.013 [−0.057, +0.033]	0.371	126	0.85	+0.046 [−0.04, +0.13]
gated token-local clamp (tuned, ours)	m2	−0.003 [−0.040, +0.030]	0.360	122	0.90	+0.053 [−0.02, +0.13]

Table 13. MMLU: all conditions, both readouts.

condition	r/o	Δacc [CI]	ECE	OCW	cr_keep	Sel [CI]
baseline	m4	—	0.113	36	—	—
Linear-AcT $\lambda=0.5$	m4	+0.010 [-0.023, +0.040]	0.112	39	0.95	+0.086 [-0.02, +0.22]
Linear-AcT $\lambda=1.0$	m4	+0.003 [-0.033, +0.040]	0.097	41	0.90	+0.095 [-0.03, +0.25]
CAST-style gate+add	m4	-0.090 [-0.137, -0.047]	0.128	15	0.60	+0.179 [+0.00, +0.36]
ablation	m4	+0.020 [-0.033, +0.070]	0.110	38	0.79	+0.239 [+0.07, +0.42]
additive -0.75	m4	-0.147 [-0.207, -0.087]	0.239	0	0.01	+0.007 [+0.00, +0.02]
clamp q_{30}	m4	+0.017 [-0.037, +0.070]	0.090	35	0.80	+0.329 [+0.14, +0.50]
clamp q_{50}	m4	+0.027 [-0.027, +0.077]	0.127	37	0.84	+0.285 [+0.12, +0.46]
clamp q_{70}	m4	+0.010 [-0.033, +0.053]	0.080	41	0.82	+0.182 [+0.01, +0.35]
gated clamp q_{50}	m4	+0.003 [-0.037, +0.040]	0.110	36	0.90	+0.178 [+0.03, +0.33]
gated OT q_{50}	m4	-0.010 [-0.033, +0.013]	0.137	35	0.97	+0.057 [-0.03, +0.16]
gated clamp q_{70}	m4	+0.030 [+0.003, +0.057]	0.139	30	0.95	+0.197 [+0.06, +0.34]
gated OT q_{70}	m4	+0.000 [-0.020, +0.020]	0.103	36	0.99	+0.070 [-0.01, +0.17]
Gaussian-OT cal.	m4	+0.013 [-0.017, +0.043]	0.099	33	0.96	+0.182 [+0.05, +0.33]
quantile-OT cal.	m4	+0.017 [-0.017, +0.047]	0.111	37	0.97	+0.134 [+0.02, +0.27]
quantile-OT non-OCW	m4	+0.037 [+0.003, +0.073]	0.101	36	0.96	+0.210 [+0.07, +0.36]
MiMiC full-space affine	m4	-0.017 [-0.070, +0.033]	0.112	57	0.83	+0.084 [-0.07, +0.25]
sequential-gate ablation	m4	+0.003 [-0.010, +0.017]	0.123	35	0.99	+0.077 [+0.00, +0.18]
sequential-gate clamp q_{50}	m4	+0.000 [+0.000, +0.000]	0.104	38	1.00	+0.000 [+0.00, +0.00]
propegate-all q_{60} ablate	m4	+0.013 [-0.030, +0.053]	0.086	36	0.89	+0.193 [+0.04, +0.36]
propegate-all q_{60} clamp q_{30}	m4	+0.010 [-0.030, +0.050]	0.076	32	0.91	+0.241 [+0.08, +0.40]
propegate-all q_{60} otq_cal	m4	+0.007 [-0.017, +0.027]	0.087	40	0.96	+0.044 [-0.04, +0.15]
propegate-all q_{75} ablate	m4	+0.017 [-0.020, +0.050]	0.111	38	0.92	+0.198 [+0.05, +0.35]
propegate-all q_{75} clamp q_{30}	m4	+0.020 [-0.017, +0.060]	0.093	31	0.95	+0.253 [+0.10, +0.41]
propegate-all q_{75} otq_cal	m4	+0.007 [-0.020, +0.033]	0.087	37	0.97	+0.050 [-0.04, +0.15]
prompting	m4	+0.007 [-0.047, +0.057]	0.107	52	0.79	+0.267 [+0.08, +0.45]
LDA-gated ablation	m4	+0.010 [-0.020, +0.037]	0.116	36	0.95	+0.197 [+0.05, +0.35]
sweep_lda-all q_{50} clamp q_{30}	m4	+0.040 [+0.010, +0.070]	0.149	33	0.93	+0.128 [-0.00, +0.28]
sweep_lda-all q_{50} clamp q_{50}	m4	+0.027 [-0.000, +0.057]	0.144	37	0.95	+0.092 [-0.01, +0.22]
gated ablation	m4	-0.010 [-0.050, +0.027]	0.085	39	0.89	+0.221 [+0.06, +0.39]
sweep_ocw-cr q_{50} clamp q_{30}	m4	+0.003 [-0.037, +0.040]	0.111	40	0.89	+0.200 [+0.04, +0.36]
gated clamp q_{50} (sweep)	m4	+0.003 [-0.037, +0.040]	0.110	36	0.90	+0.178 [+0.03, +0.33]
Linear-AcT (tuned)	m4	+0.003 [-0.033, +0.040]	0.097	41	0.90	+0.095 [-0.03, +0.25]
additive CAA (tuned)	m4	+0.003 [-0.053, +0.060]	0.084	37	0.80	+0.385 [+0.21, +0.55]
CAST-style (tuned)	m4	+0.007 [-0.020, +0.033]	0.136	32	0.93	+0.184 [+0.04, +0.34]
gated full-rank transport (tuned, ours)	m4	+0.003 [-0.037, +0.043]	0.105	49	0.91	+0.136 [-0.00, +0.29]
MiMiC (tuned)	m4	-0.013 [-0.060, +0.033]	0.092	53	0.90	+0.151 [+0.01, +0.32]
sequential gate (tuned, ours)	m4	+0.020 [-0.007, +0.050]	0.109	36	0.97	+0.022 [-0.05, +0.11]
gated local action (tuned, ours)	m4	+0.003 [-0.037, +0.043]	0.092	35	0.90	+0.290 [+0.13, +0.46]
gated k -dim transport (tuned, ours)	m4	+0.007 [-0.020, +0.033]	0.109	37	0.97	+0.078 [-0.02, +0.20]
tuned_ourpost	m4	+0.000 [+0.000, +0.000]	0.118	36	1.00	+0.000 [+0.00, +0.00]
score-proportional dose (tuned, ours)	m4	-0.003 [-0.057, +0.047]	0.084	37	0.81	+0.342 [+0.17, +0.51]
gated token-local clamp (tuned, ours)	m4	+0.043 [+0.003, +0.083]	0.129	37	0.87	+0.291 [+0.12, +0.46]
baseline	m2	—	0.242	78	—	—
Linear-AcT $\lambda=0.5$	m2	+0.007 [-0.013, +0.027]	0.230	73	0.98	+0.085 [+0.02, +0.16]
Linear-AcT $\lambda=1.0$	m2	+0.007 [-0.017, +0.030]	0.233	75	0.97	+0.075 [+0.01, +0.16]
CAST-style gate+add	m2	-0.113 [-0.160, -0.067]	0.192	89	0.74	-0.014 [-0.12, +0.10]
ablation	m2	+0.000 [-0.047, +0.043]	0.222	74	0.89	+0.233 [+0.12, +0.35]
additive -0.75	m2	-0.213 [-0.273, -0.153]	0.143	94	0.55	+0.014 [-0.11, +0.15]
clamp q_{30}	m2	-0.003 [-0.047, +0.040]	0.240	78	0.88	+0.177 [+0.07, +0.29]
clamp q_{50}	m2	-0.013 [-0.053, +0.023]	0.249	81	0.91	+0.106 [+0.01, +0.21]
clamp q_{70}	m2	+0.003 [-0.030, +0.037]	0.229	77	0.95	+0.112 [+0.02, +0.21]
gated clamp q_{50}	m2	-0.023 [-0.057, +0.003]	0.257	84	0.94	+0.026 [-0.04, +0.10]
gated OT q_{50}	m2	+0.003 [-0.017, +0.023]	0.235	76	0.98	+0.059 [+0.00, +0.13]
gated clamp q_{70}	m2	+0.000 [-0.020, +0.020]	0.237	76	0.98	+0.054 [-0.00, +0.12]
gated OT q_{70}	m2	-0.017 [-0.037, +0.000]	0.256	82	0.97	-0.002 [-0.04, +0.05]
Gaussian-OT cal.	m2	-0.007 [-0.030, +0.013]	0.241	78	0.97	+0.058 [-0.00, +0.13]
quantile-OT cal.	m2	+0.003 [-0.020, +0.023]	0.237	76	0.98	+0.067 [+0.01, +0.14]
quantile-OT non-OCW	m2	-0.010 [-0.030, +0.010]	0.242	80	0.97	+0.032 [-0.02, +0.09]
MiMiC full-space affine	m2	+0.027 [-0.010, +0.060]	0.217	68	0.95	+0.215 [+0.11, +0.33]
sequential-gate ablation	m2	+0.000 [-0.010, +0.010]	0.241	76	1.00	+0.034 [-0.00, +0.08]
sequential-gate clamp q_{50}	m2	+0.000 [-0.010, +0.010]	0.242	77	1.00	+0.021 [-0.01, +0.06]
propegate-all q_{60} ablate	m2	+0.007 [-0.030, +0.040]	0.228	74	0.94	+0.155 [+0.06, +0.26]
propegate-all q_{60} clamp q_{30}	m2	-0.007 [-0.037, +0.027]	0.234	78	0.93	+0.099 [+0.01, +0.18]
propegate-all q_{60} otq_cal	m2	+0.010 [-0.003, +0.027]	0.231	74	1.00	+0.060 [+0.01, +0.12]
propegate-all q_{75} ablate	m2	+0.027 [-0.007, +0.060]	0.219	70	0.95	+0.181 [+0.09, +0.29]
propegate-all q_{75} clamp q_{30}	m2	+0.003 [-0.027, +0.037]	0.226	76	0.95	+0.112 [+0.02, +0.20]
propegate-all q_{75} otq_cal	m2	+0.003 [-0.010, +0.017]	0.237	76	0.99	+0.042 [-0.00, +0.10]
prompting	m2	+0.010 [-0.027, +0.047]	0.231	75	0.93	+0.163 [+0.06, +0.27]
LDA-gated ablation	m2	+0.010 [-0.020, +0.037]	0.224	75	0.96	+0.105 [+0.03, +0.19]
sweep_lda-all q_{50} clamp q_{30}	m2	-0.023 [-0.057, +0.007]	0.252	84	0.93	+0.035 [-0.04, +0.12]
sweep_lda-all q_{50} clamp q_{50}	m2	-0.007 [-0.030, +0.017]	0.246	80	0.97	+0.032 [-0.02, +0.09]
gated ablation	m2	+0.000 [-0.033, +0.033]	0.229	77	0.93	+0.112 [+0.03, +0.20]
sweep_ocw-cr q_{50} clamp q_{30}	m2	-0.033 [-0.067, +0.000]	0.267	85	0.91	+0.038 [-0.04, +0.12]
gated clamp q_{50} (sweep)	m2	-0.023 [-0.057, +0.003]	0.257	84	0.94	+0.026 [-0.04, +0.10]
Linear-AcT (tuned)	m2	+0.007 [-0.017, +0.030]	0.233	75	0.97	+0.075 [+0.01, +0.16]
additive CAA (tuned)	m2	+0.027 [-0.013, +0.067]	0.191	67	0.93	+0.266 [+0.16, +0.37]
CAST-style (tuned)	m2	-0.003 [-0.030, +0.023]	0.221	75	0.95	+0.083 [+0.01, +0.16]
gated full-rank transport (tuned, ours)	m2	+0.043 [+0.017, +0.070]	0.201	64	0.99	+0.204 [+0.11, +0.30]
MiMiC (tuned)	m2	+0.013 [-0.017, +0.043]	0.229	75	0.96	+0.126 [+0.04, +0.22]
sequential gate (tuned, ours)	m2	-0.010 [-0.033, +0.010]	0.248	79	0.97	+0.045 [-0.01, +0.12]
gated local action (tuned, ours)	m2	+0.010 [-0.027, +0.043]	0.217	74	0.94	+0.167 [+0.07, +0.27]
gated k -dim transport (tuned, ours)	m2	+0.013 [-0.003, +0.033]	0.226	73	0.99	+0.081 [+0.02, +0.15]
tuned_ourpost	m2	+0.000 [+0.000, +0.000]	0.242	78	1.00	+0.000 [+0.00, +0.00]
score-proportional dose (tuned, ours)	m2	+0.020 [-0.027, +0.067]	0.207	71	0.91	+0.243 [+0.14, +0.36]
gated token-local clamp (tuned, ours)	m2	-0.003 [-0.043, +0.037]	0.243	79	0.90	+0.140 [+0.04, +0.24]

Table 14. ARC-Challenge (transfer): all conditions, both read-outs.

condition	r/o	Δacc [CI]	ECE	OCW	cr_keep	Sel [CI]
baseline	m4	—	0.258	6	—	—
Linear-AcT $\lambda=0.5$	m4	+0.043 [+0.000, +0.087]	0.297	5	0.64	−0.030 [−0.50, +0.56]
Linear-AcT $\lambda=1.0$	m4	+0.023 [−0.027, +0.070]	0.269	6	0.73	+0.227 [−0.31, +0.76]
CAST-style gate+add	m4	−0.240 [−0.313, −0.170]	0.055	0	0.00	+0.000 [+0.00, +0.00]
ablation	m4	−0.033 [−0.090, +0.023]	0.245	9	0.18	−0.151 [−0.67, +0.29]
additive −0.75	m4	−0.267 [−0.337, −0.200]	0.033	0	0.00	+0.000 [+0.00, +0.00]
clamp q_{30}	m4	−0.003 [−0.067, +0.057]	0.260	8	0.36	−0.136 [−0.67, +0.43]
clamp q_{50}	m4	+0.027 [−0.027, +0.077]	0.289	8	0.45	+0.121 [−0.46, +0.62]
clamp q_{70}	m4	+0.033 [−0.017, +0.087]	0.300	10	0.55	+0.045 [−0.50, +0.60]
gated clamp q_{50}	m4	+0.030 [−0.020, +0.080]	0.287	7	0.55	+0.212 [−0.36, +0.70]
gated OT q_{50}	m4	+0.040 [−0.007, +0.087]	0.297	5	0.55	+0.045 [−0.50, +0.62]
gated clamp q_{70}	m4	+0.017 [−0.030, +0.063]	0.279	7	0.55	+0.212 [−0.33, +0.71]
gated OT q_{70}	m4	+0.010 [−0.030, +0.047]	0.268	5	0.64	−0.030 [−0.50, +0.55]
Gaussian-OT cal.	m4	+0.020 [−0.027, +0.063]	0.280	5	0.73	+0.061 [−0.43, +0.62]
quantile-OT cal.	m4	+0.047 [+0.003, +0.093]	0.295	5	0.64	+0.136 [−0.42, +0.70]
quantile-OT non-OCW	m4	+0.003 [−0.037, +0.043]	0.257	5	0.73	+0.227 [−0.33, +0.80]
MiMiC full-space affine	m4	+0.063 [+0.007, +0.120]	0.281	11	0.45	−0.045 [−0.60, +0.50]
sequential-gate ablation	m4	+0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
proagate-all q_{60} ablate	m4	+0.007 [−0.013, +0.027]	0.262	6	0.91	+0.076 [−0.25, +0.50]
proagate-all q_{60} clamp q_{30}	m4	+0.000 [−0.017, +0.017]	0.256	8	1.00	+0.000 [+0.00, +0.00]
proagate-all q_{60} otq_cal	m4	+0.010 [−0.003, +0.023]	0.257	6	1.00	+0.167 [+0.00, +0.50]
proagate-all q_{75} ablate	m4	+0.007 [−0.007, +0.020]	0.262	6	0.91	+0.076 [−0.25, +0.50]
proagate-all q_{75} clamp q_{30}	m4	+0.003 [−0.007, +0.013]	0.262	6	1.00	+0.167 [+0.00, +0.50]
proagate-all q_{75} otq_cal	m4	+0.000 [−0.010, +0.010]	0.257	6	1.00	+0.167 [+0.00, +0.50]
prompting	m4	−0.030 [−0.090, +0.033]	0.195	15	0.27	+0.106 [−0.36, +0.50]
LDA-gated ablation	m4	+0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
sweep_lda-all q_{50} clamp q_{30}	m4	+0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
sweep_lda-all q_{50} clamp q_{50}	m4	+0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
gated ablation	m4	−0.050 [−0.107, +0.010]	0.235	11	0.18	−0.318 [−0.86, +0.22]
sweep_ocwcr-cr q_{50} clamp q_{30}	m4	−0.007 [−0.067, +0.047]	0.267	10	0.36	−0.303 [−0.79, +0.25]
gated clamp q_{50} (sweep)	m4	+0.030 [−0.020, +0.080]	0.287	7	0.55	+0.212 [−0.36, +0.70]
Linear-AcT (tuned)	m4	+0.023 [−0.027, +0.070]	0.269	6	0.73	+0.227 [−0.31, +0.76]
additive CAA (tuned)	m4	−0.030 [−0.093, +0.027]	0.243	1	0.00	+0.000 [+0.00, +0.00]
CAST-style (tuned)	m4	−0.063 [−0.120, −0.010]	0.218	7	0.27	+0.106 [−0.33, +0.50]
gated full-rank transport (tuned, ours)	m4	+0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
MiMiC (tuned)	m4	−0.013 [−0.070, +0.040]	0.233	11	0.55	+0.045 [−0.47, +0.59]
sequential gate (tuned, ours)	m4	+0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
gated local action (tuned, ours)	m4	+0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
gated k -dim transport (tuned, ours)	m4	+0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
tuned_ourspost	m4	+0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
score-proportional dose (tuned, ours)	m4	+0.003 [−0.007, +0.013]	0.259	6	1.00	+0.000 [+0.00, +0.00]
gated token-local clamp (tuned, ours)	m4	+0.000 [+0.000, +0.000]	0.258	6	1.00	+0.000 [+0.00, +0.00]
baseline	m2	—	0.216	87	—	—
Linear-AcT $\lambda=0.5$	m2	+0.013 [−0.017, +0.047]	0.204	85	0.94	+0.092 [+0.01, +0.18]
Linear-AcT $\lambda=1.0$	m2	+0.020 [−0.010, +0.053]	0.202	87	0.95	+0.121 [+0.04, +0.21]
CAST-style gate+add	m2	−0.280 [−0.347, −0.217]	0.226	102	0.30	−0.107 [−0.23, +0.01]
ablation	m2	+0.030 [−0.020, +0.077]	0.171	70	0.88	+0.335 [+0.22, +0.45]
additive −0.75	m2	−0.290 [−0.360, −0.220]	0.218	105	0.26	−0.081 [−0.20, +0.03]
clamp q_{30}	m2	+0.040 [+0.000, +0.073]	0.192	71	0.94	+0.283 [+0.18, +0.38]
clamp q_{50}	m2	+0.013 [−0.033, +0.057]	0.213	74	0.89	+0.288 [+0.17, +0.40]
clamp q_{70}	m2	+0.030 [−0.010, +0.070]	0.192	69	0.94	+0.283 [+0.18, +0.39]
gated clamp q_{50}	m2	+0.030 [−0.010, +0.070]	0.199	78	0.91	+0.222 [+0.12, +0.33]
gated OT q_{50}	m2	+0.023 [−0.013, +0.060]	0.191	76	0.94	+0.168 [+0.07, +0.26]
gated clamp q_{70}	m2	+0.030 [−0.010, +0.067]	0.196	77	0.92	+0.181 [+0.09, +0.28]
gated OT q_{70}	m2	+0.017 [−0.017, +0.050]	0.205	82	0.93	+0.128 [+0.04, +0.22]
Gaussian-OT cal.	m2	−0.010 [−0.043, +0.027]	0.224	91	0.92	+0.055 [−0.02, +0.14]
quantile-OT cal.	m2	+0.003 [−0.027, +0.033]	0.201	80	0.94	+0.138 [+0.05, +0.23]
quantile-OT non-OCW	m2	+0.000 [−0.037, +0.033]	0.212	85	0.92	+0.078 [−0.00, +0.17]
MiMiC full-space affine	m2	+0.030 [−0.020, +0.080]	0.194	77	0.87	+0.307 [+0.19, +0.42]
sequential-gate ablation	m2	+0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]
proagate-all q_{60} ablate	m2	−0.007 [−0.027, +0.010]	0.220	89	0.97	+0.003 [−0.04, +0.05]
proagate-all q_{60} clamp q_{30}	m2	+0.003 [−0.010, +0.017]	0.212	86	0.98	+0.019 [−0.02, +0.07]
proagate-all q_{60} otq_cal	m2	+0.003 [−0.013, +0.020]	0.216	86	0.98	+0.030 [−0.01, +0.08]
proagate-all q_{75} ablate	m2	−0.003 [−0.020, +0.010]	0.222	91	0.98	−0.004 [−0.03, +0.03]
proagate-all q_{75} clamp q_{30}	m2	+0.013 [+0.003, +0.027]	0.208	86	0.99	+0.006 [−0.01, +0.03]
proagate-all q_{75} otq_cal	m2	−0.010 [−0.023, +0.000]	0.224	90	0.98	−0.004 [−0.03, +0.03]
prompting	m2	+0.010 [−0.043, +0.063]	0.226	88	0.83	+0.191 [+0.08, +0.31]
LDA-gated ablation	m2	+0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]
sweep_lda-all q_{50} clamp q_{30}	m2	+0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]
sweep_lda-all q_{50} clamp q_{50}	m2	+0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]
gated ablation	m2	+0.013 [−0.040, +0.063]	0.190	76	0.84	+0.258 [+0.14, +0.37]
sweep_ocwcr-cr q_{50} clamp q_{30}	m2	+0.043 [+0.003, +0.083]	0.179	72	0.93	+0.295 [+0.19, +0.40]
gated clamp q_{50} (sweep)	m2	+0.030 [−0.010, +0.070]	0.199	78	0.91	+0.222 [+0.12, +0.33]
Linear-AcT (tuned)	m2	+0.020 [−0.010, +0.053]	0.202	87	0.95	+0.121 [+0.04, +0.21]
additive CAA (tuned)	m2	+0.023 [−0.030, +0.077]	0.133	69	0.80	+0.343 [+0.23, +0.46]
CAST-style (tuned)	m2	−0.060 [−0.110, −0.013]	0.204	77	0.76	+0.182 [+0.05, +0.30]
gated full-rank transport (tuned, ours)	m2	+0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]
MiMiC (tuned)	m2	−0.007 [−0.053, +0.040]	0.249	91	0.86	+0.147 [+0.04, +0.26]
sequential gate (tuned, ours)	m2	+0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]
gated local action (tuned, ours)	m2	+0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]
gated k -dim transport (tuned, ours)	m2	+0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]
tuned_ourspost	m2	+0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]
score-proportional dose (tuned, ours)	m2	+0.000 [−0.010, +0.010]	0.217	85	0.99	+0.013 [−0.02, +0.05]
gated token-local clamp (tuned, ours)	m2	+0.000 [+0.000, +0.000]	0.216	87	1.00	+0.000 [+0.00, +0.00]

Table 15. GSM8K-MCQ (transfer): all conditions, both read-outs. Note the tiny OCW/CR poles (6/12 of 300 at baseline); the model is underconfident on this domain, selectivity CIs are wide, and transport maps correctly act near-identity.

method	HumanEval	WikiText	open-ended (AlpacaEval prompts)			gate fires
	pass@1	ppl	fluency ppl	distinct-3	win rate	
unsteered	0.396	23.4	3.00	0.953	–	–
additive -0.75 [20]	0.128	47.3	3.36	0.480	0.043	–
additive, tuned dose [20]	0.408	23.6	3.14	0.922	0.495	–
MiMiC [24]	0.396	24.2	3.01	0.949	0.512	–
MiMiC, tuned layer [24]	0.384	24.6	3.02	0.948	0.458	–
Linear-AcT [21]	0.390	23.4	2.98	0.950	0.505	–
clamp q_{50} (ours)	0.390	23.1	2.97	0.946	0.507	–
quantile-OT (ours)	0.396	23.3	2.99	0.951	0.492	–
online-gated ablation (ours)	0.396	n/a	3.01	0.946	0.477	0.680
gated full-rank transport (ours)	0.396	n/a	3.02	0.954	0.470	0.586

Table 16. Capability preservation away from the fitted behavior. The unconditional rows keep the action on for every token, so they bound what the gated operator can cost. WikiText is scored teacher-forced, which gives a sequential gate no generated prefix to read, so that cell is not applicable. The win rate is order-consistent against the unsteered answer, judged by Qwen2.5-7B-Instruct in both presentation orders with ties counted as half, so 0.5 means indistinguishable.

operator	MMLU				ARC				GSM8K			
	disp.	NN	out%	KL	disp.	NN	out%	KL	disp.	NN	out%	KL
random direction, norm-matched	0.663	1.90	1.6	0.445	0.644	2.27	1.7	0.233	0.749	2.06	0.9	0.149
additive -0.75 [20]	0.663	1.88	100.0	0.032	0.644	2.25	100.0	0.008	0.749	2.04	100.0	0.012
additive, tuned dose [20]	0.221	1.22	69.9	0.005	0.215	1.33	40.3	0.001	–	–	–	–
MiMiC [24]	0.082	1.06	1.3	0.001	0.074	1.08	1.8	0.000	0.097	1.06	4.5	0.001
Linear-AcT [21]	0.012	1.01	1.0	0.000	0.011	1.01	1.5	0.000	0.012	1.01	3.5	0.000
clamp q_{50} (ours)	0.034	1.03	1.3	0.006	0.046	1.06	1.1	0.001	0.003	1.00	2.9	0.001
quantile-OT (ours)	0.004	1.00	1.0	0.000	0.003	1.00	1.3	0.000	0.007	1.00	4.0	0.000
ablation [1]	0.144	1.15	29.5	0.010	0.178	1.28	28.4	0.001	0.062	1.05	10.1	0.001

Table 17. On-manifold diagnostics at layer 14. NN is the mean nearest-neighbour distance to a bank of natural activations relative to the unsteered value, so 1.00 means the edit lands as close to the model’s own states as the states it replaced; out% is the fraction of tokens outside the 99% conditional ellipsoid, against 1% expected; KL is the mean divergence of the next-token distribution, with 0.1 the accepted threshold [1].

method	passes	seconds	wall clock
unsteered	1	43.5	1.00×
additive -0.75 [20]	1	45.2	1.04×
ablation [1]	1	44.6	1.02×
clamp q_{50} (ours)	1	41.4	0.95×
quantile-OT (ours)	1	44.8	1.03×
MiMiC [24]	1	42.9	0.98×
Linear-Act [21]	1	40.9	0.94×
gated ablation, post-hoc gate (ours) (223/300 regenerated)	2	85.8	1.97×
gated ablation, sequential gate (ours)	1	45.5	1.05×

Table 18. End-to-end wall clock, MMLU $n=300$, batch 64, one H100.

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стиринг активаций and оптимальный транспорт and калибровка and
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